

# ESCALATOR SCHEMATIC/WIRING DIAGRAM

|                 |   |      |
|-----------------|---|------|
| PROJ.NAME       | : |      |
| PROJ.NO         | : |      |
| ESCALATOR TYPE  | : | S-BT |
| ESCALATOR MODEL | : |      |
| ESCALATOR NO    | : |      |
| POWER SUPPLY    | : |      |
| CONTROL SYSTEM  | : |      |

 **HYUNDAI ELEVATOR CO., LTD.**

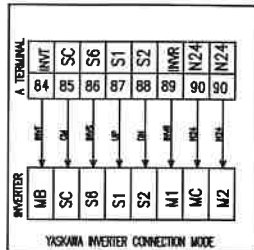
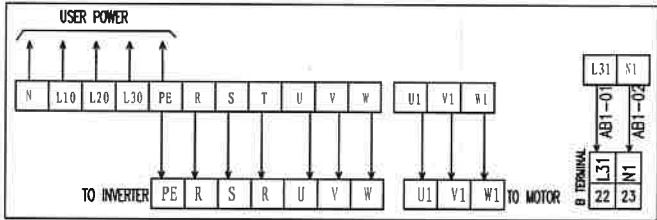
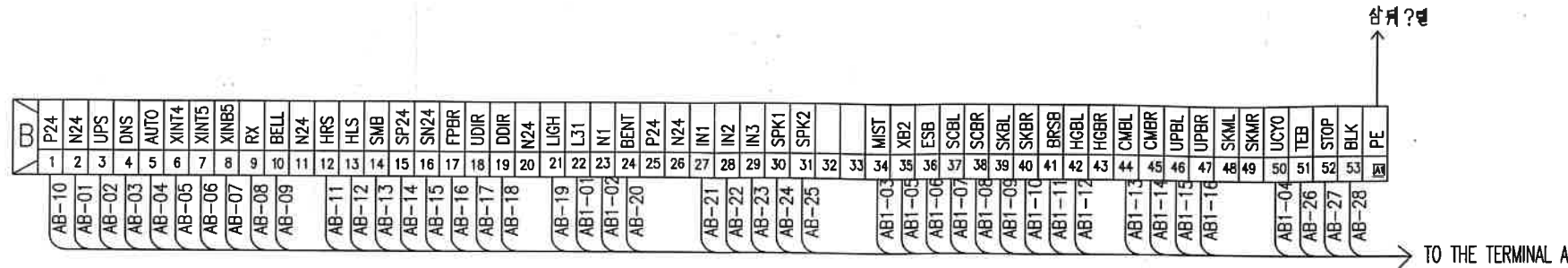
도  
면  
부  
호

CHANGE  
개정판

시 ADD PE TERMINAL(C.9)

H.F.F 2018.10.11

COL 2



| SIZE | WIRE  | TERMINAL | WIRE  | TERMINAL | WIRE  | TERMINAL |
|------|-------|----------|-------|----------|-------|----------|
| 1    | 1.5   | 1.5      | 1.5   | 1.5      | 1.5   | 1.5      |
| 2    | 2.5   | 2.5      | 2.5   | 2.5      | 2.5   | 2.5      |
| 3    | 3.5   | 3.5      | 3.5   | 3.5      | 3.5   | 3.5      |
| 4    | 4.5   | 4.5      | 4.5   | 4.5      | 4.5   | 4.5      |
| 5    | 5.5   | 5.5      | 5.5   | 5.5      | 5.5   | 5.5      |
| 6    | 6.5   | 6.5      | 6.5   | 6.5      | 6.5   | 6.5      |
| 7    | 7.5   | 7.5      | 7.5   | 7.5      | 7.5   | 7.5      |
| 8    | 8.5   | 8.5      | 8.5   | 8.5      | 8.5   | 8.5      |
| 9    | 9.5   | 9.5      | 9.5   | 9.5      | 9.5   | 9.5      |
| 10   | 10.5  | 10.5     | 10.5  | 10.5     | 10.5  | 10.5     |
| 11   | 11.5  | 11.5     | 11.5  | 11.5     | 11.5  | 11.5     |
| 12   | 12.5  | 12.5     | 12.5  | 12.5     | 12.5  | 12.5     |
| 13   | 13.5  | 13.5     | 13.5  | 13.5     | 13.5  | 13.5     |
| 14   | 14.5  | 14.5     | 14.5  | 14.5     | 14.5  | 14.5     |
| 15   | 15.5  | 15.5     | 15.5  | 15.5     | 15.5  | 15.5     |
| 16   | 16.5  | 16.5     | 16.5  | 16.5     | 16.5  | 16.5     |
| 17   | 17.5  | 17.5     | 17.5  | 17.5     | 17.5  | 17.5     |
| 18   | 18.5  | 18.5     | 18.5  | 18.5     | 18.5  | 18.5     |
| 19   | 19.5  | 19.5     | 19.5  | 19.5     | 19.5  | 19.5     |
| 20   | 20.5  | 20.5     | 20.5  | 20.5     | 20.5  | 20.5     |
| 21   | 21.5  | 21.5     | 21.5  | 21.5     | 21.5  | 21.5     |
| 22   | 22.5  | 22.5     | 22.5  | 22.5     | 22.5  | 22.5     |
| 23   | 23.5  | 23.5     | 23.5  | 23.5     | 23.5  | 23.5     |
| 24   | 24.5  | 24.5     | 24.5  | 24.5     | 24.5  | 24.5     |
| 25   | 25.5  | 25.5     | 25.5  | 25.5     | 25.5  | 25.5     |
| 26   | 26.5  | 26.5     | 26.5  | 26.5     | 26.5  | 26.5     |
| 27   | 27.5  | 27.5     | 27.5  | 27.5     | 27.5  | 27.5     |
| 28   | 28.5  | 28.5     | 28.5  | 28.5     | 28.5  | 28.5     |
| 29   | 29.5  | 29.5     | 29.5  | 29.5     | 29.5  | 29.5     |
| 30   | 30.5  | 30.5     | 30.5  | 30.5     | 30.5  | 30.5     |
| 31   | 31.5  | 31.5     | 31.5  | 31.5     | 31.5  | 31.5     |
| 32   | 32.5  | 32.5     | 32.5  | 32.5     | 32.5  | 32.5     |
| 33   | 33.5  | 33.5     | 33.5  | 33.5     | 33.5  | 33.5     |
| 34   | 34.5  | 34.5     | 34.5  | 34.5     | 34.5  | 34.5     |
| 35   | 35.5  | 35.5     | 35.5  | 35.5     | 35.5  | 35.5     |
| 36   | 36.5  | 36.5     | 36.5  | 36.5     | 36.5  | 36.5     |
| 37   | 37.5  | 37.5     | 37.5  | 37.5     | 37.5  | 37.5     |
| 38   | 38.5  | 38.5     | 38.5  | 38.5     | 38.5  | 38.5     |
| 39   | 39.5  | 39.5     | 39.5  | 39.5     | 39.5  | 39.5     |
| 40   | 40.5  | 40.5     | 40.5  | 40.5     | 40.5  | 40.5     |
| 41   | 41.5  | 41.5     | 41.5  | 41.5     | 41.5  | 41.5     |
| 42   | 42.5  | 42.5     | 42.5  | 42.5     | 42.5  | 42.5     |
| 43   | 43.5  | 43.5     | 43.5  | 43.5     | 43.5  | 43.5     |
| 44   | 44.5  | 44.5     | 44.5  | 44.5     | 44.5  | 44.5     |
| 45   | 45.5  | 45.5     | 45.5  | 45.5     | 45.5  | 45.5     |
| 46   | 46.5  | 46.5     | 46.5  | 46.5     | 46.5  | 46.5     |
| 47   | 47.5  | 47.5     | 47.5  | 47.5     | 47.5  | 47.5     |
| 48   | 48.5  | 48.5     | 48.5  | 48.5     | 48.5  | 48.5     |
| 49   | 49.5  | 49.5     | 49.5  | 49.5     | 49.5  | 49.5     |
| 50   | 50.5  | 50.5     | 50.5  | 50.5     | 50.5  | 50.5     |
| 51   | 51.5  | 51.5     | 51.5  | 51.5     | 51.5  | 51.5     |
| 52   | 52.5  | 52.5     | 52.5  | 52.5     | 52.5  | 52.5     |
| 53   | 53.5  | 53.5     | 53.5  | 53.5     | 53.5  | 53.5     |
| 54   | 54.5  | 54.5     | 54.5  | 54.5     | 54.5  | 54.5     |
| 55   | 55.5  | 55.5     | 55.5  | 55.5     | 55.5  | 55.5     |
| 56   | 56.5  | 56.5     | 56.5  | 56.5     | 56.5  | 56.5     |
| 57   | 57.5  | 57.5     | 57.5  | 57.5     | 57.5  | 57.5     |
| 58   | 58.5  | 58.5     | 58.5  | 58.5     | 58.5  | 58.5     |
| 59   | 59.5  | 59.5     | 59.5  | 59.5     | 59.5  | 59.5     |
| 60   | 60.5  | 60.5     | 60.5  | 60.5     | 60.5  | 60.5     |
| 61   | 61.5  | 61.5     | 61.5  | 61.5     | 61.5  | 61.5     |
| 62   | 62.5  | 62.5     | 62.5  | 62.5     | 62.5  | 62.5     |
| 63   | 63.5  | 63.5     | 63.5  | 63.5     | 63.5  | 63.5     |
| 64   | 64.5  | 64.5     | 64.5  | 64.5     | 64.5  | 64.5     |
| 65   | 65.5  | 65.5     | 65.5  | 65.5     | 65.5  | 65.5     |
| 66   | 66.5  | 66.5     | 66.5  | 66.5     | 66.5  | 66.5     |
| 67   | 67.5  | 67.5     | 67.5  | 67.5     | 67.5  | 67.5     |
| 68   | 68.5  | 68.5     | 68.5  | 68.5     | 68.5  | 68.5     |
| 69   | 69.5  | 69.5     | 69.5  | 69.5     | 69.5  | 69.5     |
| 70   | 70.5  | 70.5     | 70.5  | 70.5     | 70.5  | 70.5     |
| 71   | 71.5  | 71.5     | 71.5  | 71.5     | 71.5  | 71.5     |
| 72   | 72.5  | 72.5     | 72.5  | 72.5     | 72.5  | 72.5     |
| 73   | 73.5  | 73.5     | 73.5  | 73.5     | 73.5  | 73.5     |
| 74   | 74.5  | 74.5     | 74.5  | 74.5     | 74.5  | 74.5     |
| 75   | 75.5  | 75.5     | 75.5  | 75.5     | 75.5  | 75.5     |
| 76   | 76.5  | 76.5     | 76.5  | 76.5     | 76.5  | 76.5     |
| 77   | 77.5  | 77.5     | 77.5  | 77.5     | 77.5  | 77.5     |
| 78   | 78.5  | 78.5     | 78.5  | 78.5     | 78.5  | 78.5     |
| 79   | 79.5  | 79.5     | 79.5  | 79.5     | 79.5  | 79.5     |
| 80   | 80.5  | 80.5     | 80.5  | 80.5     | 80.5  | 80.5     |
| 81   | 81.5  | 81.5     | 81.5  | 81.5     | 81.5  | 81.5     |
| 82   | 82.5  | 82.5     | 82.5  | 82.5     | 82.5  | 82.5     |
| 83   | 83.5  | 83.5     | 83.5  | 83.5     | 83.5  | 83.5     |
| 84   | 84.5  | 84.5     | 84.5  | 84.5     | 84.5  | 84.5     |
| 85   | 85.5  | 85.5     | 85.5  | 85.5     | 85.5  | 85.5     |
| 86   | 86.5  | 86.5     | 86.5  | 86.5     | 86.5  | 86.5     |
| 87   | 87.5  | 87.5     | 87.5  | 87.5     | 87.5  | 87.5     |
| 88   | 88.5  | 88.5     | 88.5  | 88.5     | 88.5  | 88.5     |
| 89   | 89.5  | 89.5     | 89.5  | 89.5     | 89.5  | 89.5     |
| 90   | 90.5  | 90.5     | 90.5  | 90.5     | 90.5  | 90.5     |
| 91   | 91.5  | 91.5     | 91.5  | 91.5     | 91.5  | 91.5     |
| 92   | 92.5  | 92.5     | 92.5  | 92.5     | 92.5  | 92.5     |
| 93   | 93.5  | 93.5     | 93.5  | 93.5     | 93.5  | 93.5     |
| 94   | 94.5  | 94.5     | 94.5  | 94.5     | 94.5  | 94.5     |
| 95   | 95.5  | 95.5     | 95.5  | 95.5     | 95.5  | 95.5     |
| 96   | 96.5  | 96.5     | 96.5  | 96.5     | 96.5  | 96.5     |
| 97   | 97.5  | 97.5     | 97.5  | 97.5     | 97.5  | 97.5     |
| 98   | 98.5  | 98.5     | 98.5  | 98.5     | 98.5  | 98.5     |
| 99   | 99.5  | 99.5     | 99.5  | 99.5     | 99.5  | 99.5     |
| 100  | 100.5 | 100.5    | 100.5 | 100.5    | 100.5 | 100.5    |

HYUNDAI  
ELEVATOR CO., LTD.

표  
면  
제  
도  
설  
계  
검  
토  
승  
인  
H.F.F  
2016.11.1  
成東興  
2018.10.11  
倪偉博  
2018.10.11

UNIT  
단위  
SCALE  
척도  
3RD ANGLE  
3 각 법  
REF. NO  
참고도면

APPAR. PLC  
TITLE CONTROL TRUSS WIRING  
DWG. NO CHE000050

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|

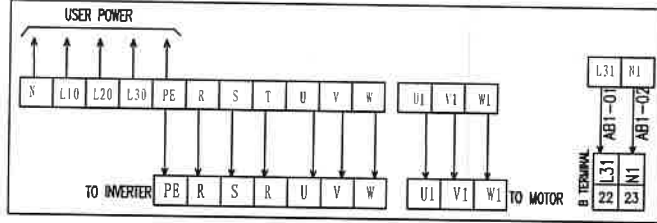
 $\frac{2}{2}$

| REV | DATE | DESCRIPTION |
|-----|------|-------------|
| 1   |      |             |
| 2   |      |             |
| 3   |      |             |
| 4   |      |             |
| 5   |      |             |
| 6   |      |             |
| 7   |      |             |
| 8   |      |             |
| 9   |      |             |
| 10  |      |             |

CHANGE  
개 정 란

ADD PE TERMINAL(C9)

H.FF 2018.10.11



| TOLERANCES-UNLESS OTHERWISE SPECIFIED |               |             |                      |                      |              |
|---------------------------------------|---------------|-------------|----------------------|----------------------|--------------|
| SIZE                                  | FRONT<br>DIM. | TOP<br>DIM. | MAX.<br>DIM.<br>TOL. | MIN.<br>DIM.<br>TOL. | FIN.<br>DIM. |
| 1 4                                   | ± 0.1         | ± 0.2       | ± 0.3                | ± 0.5                | ± 0.5        |
| 4 16                                  | ± 0.2         | ± 0.3       | ± 0.5                | ± 0.7                | ± 0.7        |
| 16 40                                 | ± 0.3         | ± 0.4       | ± 0.7                | ± 0.9                | ± 0.9        |
| 40 100                                | ± 0.4         | ± 0.5       | ± 1.2                | ± 1.5                | ± 1.5        |
| 100 200                               | ± 0.5         | ± 0.6       | ± 1.5                | ± 1.8                | ± 1.8        |
| 200 500                               | ± 0.6         | ± 0.8       | ± 2.0                | ± 2.5                | ± 2.5        |
| 500 1000                              | ± 0.8         | ± 1.0       | ± 2.5                | ± 3.0                | ± 3.0        |
| 1000 2000                             | ± 1.0         | ± 1.5       | ± 3.0                | ± 3.5                | ± 3.5        |
| FIN.                                  | ± 0.1         | ± 0.2       | ± 0.3                | ± 0.5                | ± 0.5        |

**HYUNDAI**  
ELEVATOR CO., LTD.

| 표 준 | DRWN BY    | DSND BY    | CHKD BY    | APPD BY |
|-----|------------|------------|------------|---------|
| 제 도 | H.FF       | 김 보        | 김 보        | 김 보     |
| 임 시 |            |            |            |         |
| 보 류 | 2017.03.08 | 2018.10.11 | 2018.10.11 |         |

| UNIT    | MM.     | APPAR.                   |
|---------|---------|--------------------------|
| 단 위     | 1       | PLC                      |
| SCALE   | 1       | TITLE                    |
| 척 도     | 3 각 법   | 제 목 CONTROL TRUSS WIRING |
| REF. NO | DWG. NO |                          |
| 참 고 도 면 | 도면번호    | CHE000054                |

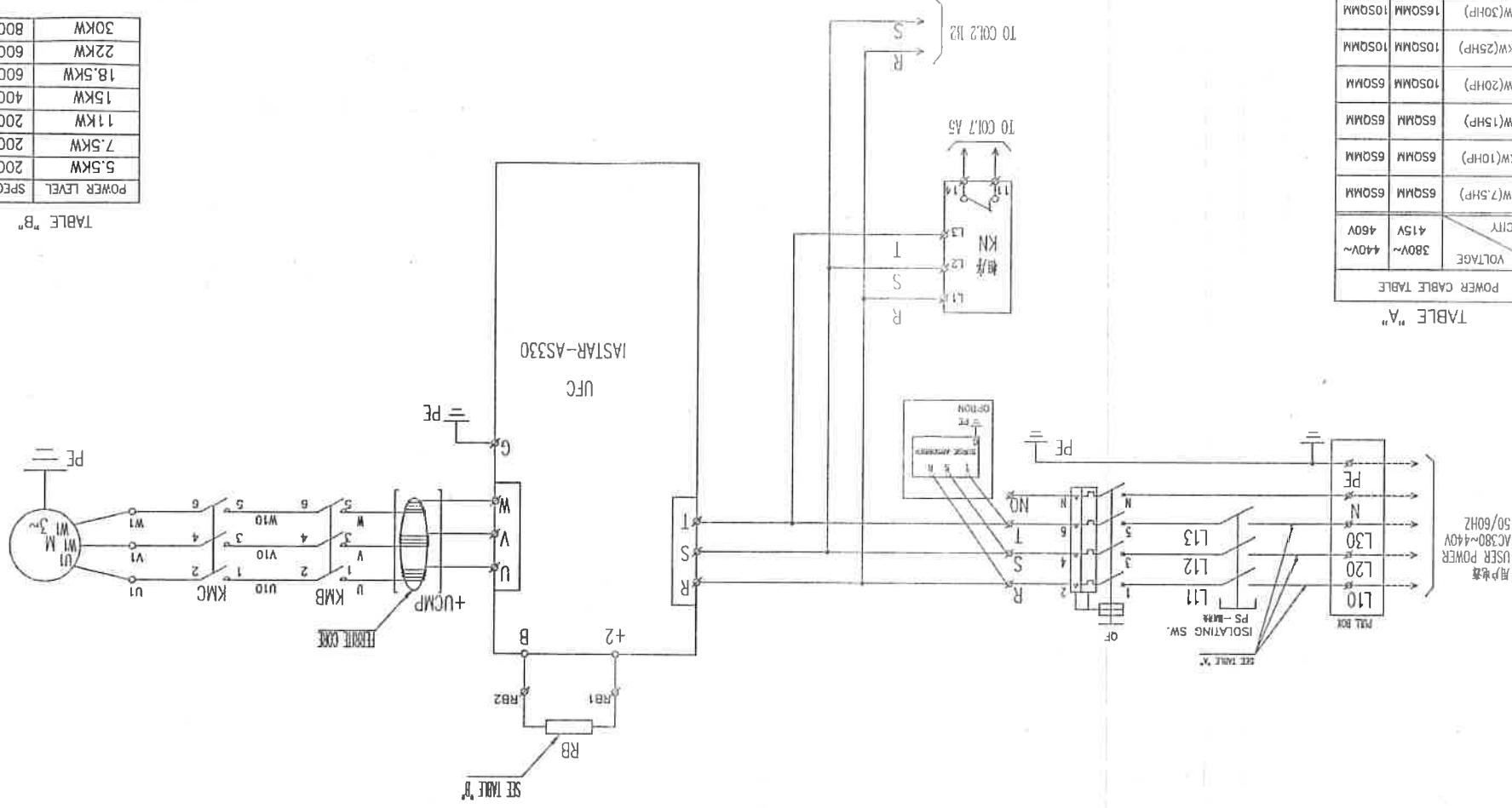
COL 2

| 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10     |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| AB-10 | AB-01 | AB-02 | AB-03 | AB-04 | AB-05 | AB-06 | AB-07 | AB-08 | AB-09  |
| AB-11 | AB-12 | AB-13 | AB-14 | AB-15 | AB-16 | AB-17 | AB-18 | AB-19 | AB-20  |
| AB-21 | AB-22 | AB-23 | AB-24 | AB-25 | AB-26 | AB-27 | AB-28 | AB-29 | AB-30  |
| AB-31 | AB-32 | AB-33 | AB-34 | AB-35 | AB-36 | AB-37 | AB-38 | AB-39 | AB-40  |
| AB-41 | AB-42 | AB-43 | AB-44 | AB-45 | AB-46 | AB-47 | AB-48 | AB-49 | AB-50  |
| AB-51 | AB-52 | AB-53 | AB-54 | AB-55 | AB-56 | AB-57 | AB-58 | AB-59 | AB-60  |
| AB-61 | AB-62 | AB-63 | AB-64 | AB-65 | AB-66 | AB-67 | AB-68 | AB-69 | AB-70  |
| AB-71 | AB-72 | AB-73 | AB-74 | AB-75 | AB-76 | AB-77 | AB-78 | AB-79 | AB-80  |
| AB-81 | AB-82 | AB-83 | AB-84 | AB-85 | AB-86 | AB-87 | AB-88 | AB-89 | AB-90  |
| AB-91 | AB-92 | AB-93 | AB-94 | AB-95 | AB-96 | AB-97 | AB-98 | AB-99 | AB-100 |

삼선?별

TO THE TERMINAL A

|                   |              |
|-------------------|--------------|
| POWER CABLE TABLE |              |
| VOLTAGE           | CAPACITY     |
| 380V~440V         | 415V         |
|                   | 460V         |
| 650MM             | 5.5KW(7.5HP) |
| 650MM             | 7.5KW(10HP)  |
| 650MM             | 11KW(15HP)   |
| 650MM             | 15KW(20HP)   |
| 1050MM            | 18.5KW(25HP) |
| 1050MM            | 22KW(30HP)   |
| 1650MM            | 30KW(40HP)   |
| 2550MM            | 37KW(50HP)   |



NOTE: THIS DRAWING APPLIED TO INVERTER.

[illegible]

|                                                                                                                        |               |               |               |              |
|------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|--------------|
|  <b>HYUNDAI</b><br>ELEVATOR CO., LTD. | DMMN BY<br>제도 | DSND BY<br>설계 | CHKD BY<br>검토 | APD BY<br>승인 |
|                                                                                                                        | 위치            | 층도            | 시공            | 승인           |
| H.F.F                                                                                                                  |               | 2017.05.10    |               | 날짜           |

|           |         |
|-----------|---------|
| UNIT      | MM.     |
| SCALE     | 1 / 1   |
| 3RD ANGLE | 3 2 1   |
| REF. NO   | REF. NO |

| APPAR. | TIME | 제목 | DWG. NO |
|--------|------|----|---------|
|        |      |    |         |

MCU

IRCUIT SCHEMATIC

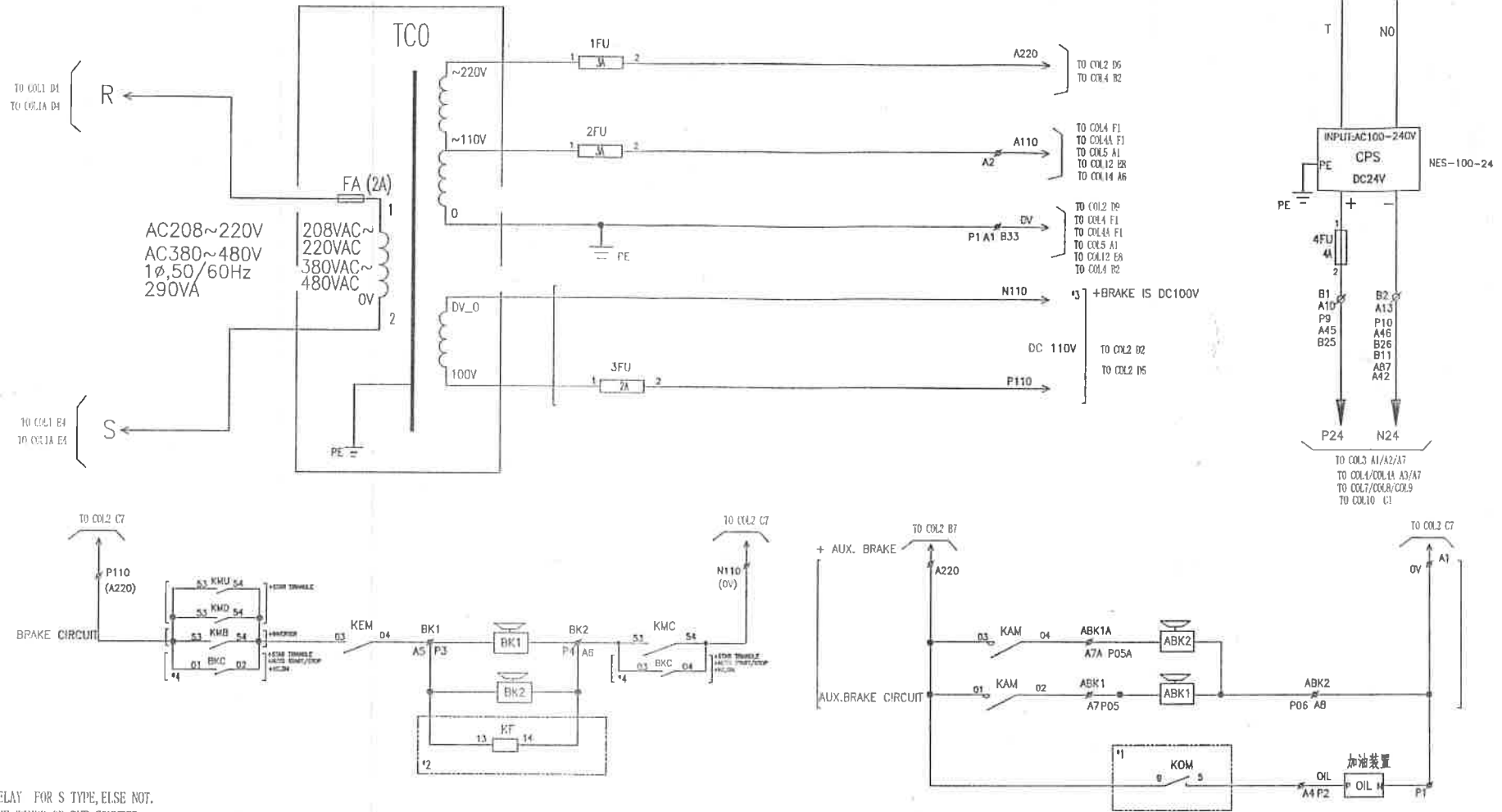
22 MAY 2017

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
해기없이 복사할 수 없음

| REV | DATE | DESCRIPTION |
|-----|------|-------------|
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |

NOTE:  
\*1:ADDED KOM RELAY FOR S TYPE, ELSE NOT.  
\*2:ADDED FOR THE TIMER OR THE COUNTER .  
\*3:ADDED FOR THE VOLTAGE DC110V OF THE MOTOR BRAKE.  
\*4:USED IF PROVIDED WITH MULTI-POST OR AUTO START/STOP(EXCEPT CHINA)

COL 2



图纸确认章  
22 MAY 2017

| REV | DATE       | DESCRIPTION |
|-----|------------|-------------|
| 1   | 2017.05.10 |             |
| 2   | 2017.05.10 |             |
| 3   | 2017.05.10 |             |
| 4   | 2017.05.10 |             |
| 5   | 2017.05.10 |             |
| 6   | 2017.05.10 |             |
| 7   | 2017.05.10 |             |
| 8   | 2017.05.10 |             |
| 9   | 2017.05.10 |             |
| 10  | 2017.05.10 |             |

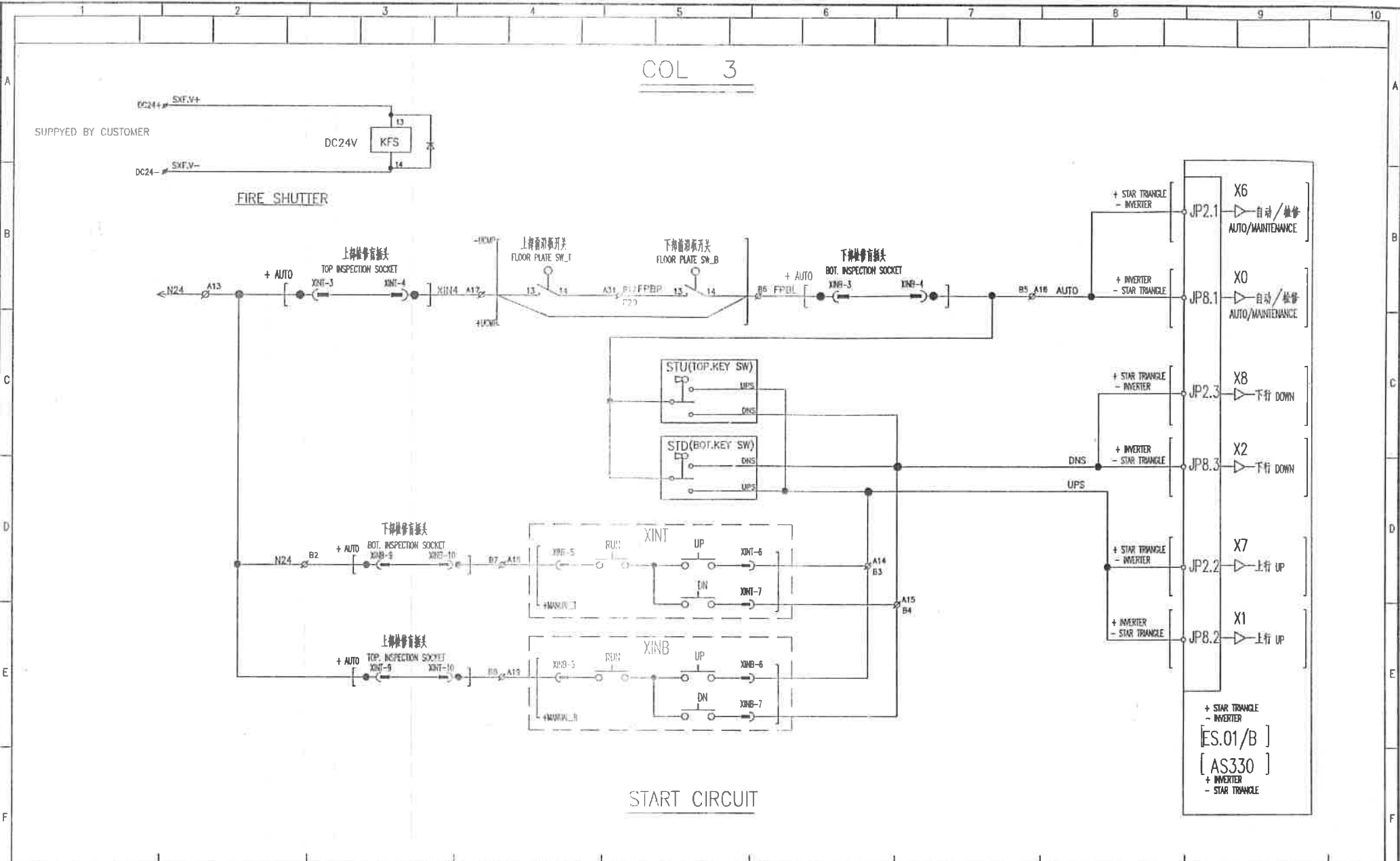
HYUNDAI  
ELEVATOR CO., LTD

| REV | DATE       | DESCRIPTION |
|-----|------------|-------------|
| 1   | 2017.05.10 |             |
| 2   | 2017.05.10 |             |
| 3   | 2017.05.10 |             |
| 4   | 2017.05.10 |             |
| 5   | 2017.05.10 |             |
| 6   | 2017.05.10 |             |
| 7   | 2017.05.10 |             |
| 8   | 2017.05.10 |             |
| 9   | 2017.05.10 |             |
| 10  | 2017.05.10 |             |

| UNIT      | MM    | APPAR.  | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면  | 도면번호    | 2/19                      |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
해기점이 복사할 수 없음

| REV | DATE | DESCRIPTION |
|-----|------|-------------|
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |
|     |      |             |



START CIRCUIT

图纸确认  
27 MAY 2017

| NO. | REV. | DATE       | DESCRIPTION |
|-----|------|------------|-------------|
| 1   | 1    | 2017.05.10 | INITIAL     |
| 2   | 2    | 2017.05.10 | REVISED     |
| 3   | 3    | 2017.05.10 | REVISED     |
| 4   | 4    | 2017.05.10 | REVISED     |
| 5   | 5    | 2017.05.10 | REVISED     |
| 6   | 6    | 2017.05.10 | REVISED     |
| 7   | 7    | 2017.05.10 | REVISED     |
| 8   | 8    | 2017.05.10 | REVISED     |
| 9   | 9    | 2017.05.10 | REVISED     |
| 10  | 10   | 2017.05.10 | REVISED     |

**HYUNDAI**  
ELEVATOR CO., LTD.

표준 제도 설계 검토 승인  
임시 H.FF  
보류 2017.05.10

| UNIT      | MM.   | APPAR.  | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면  | DWG. NO | 도면번호                      |





[illegible]

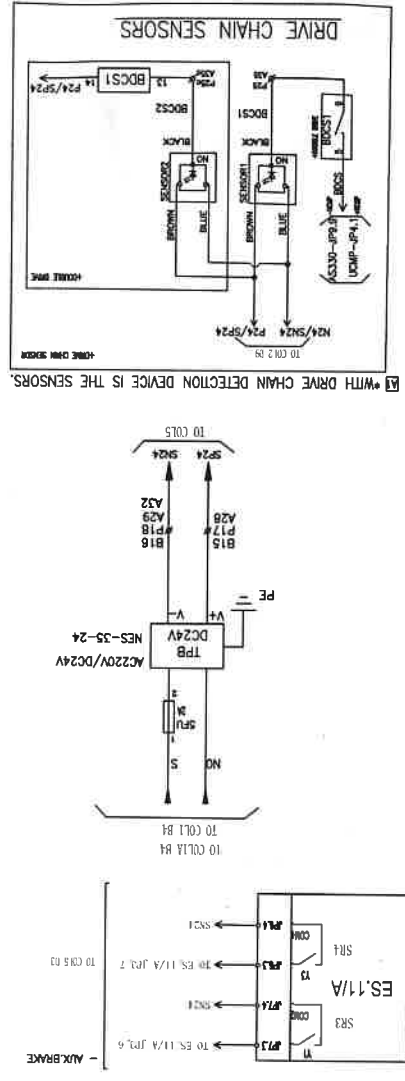
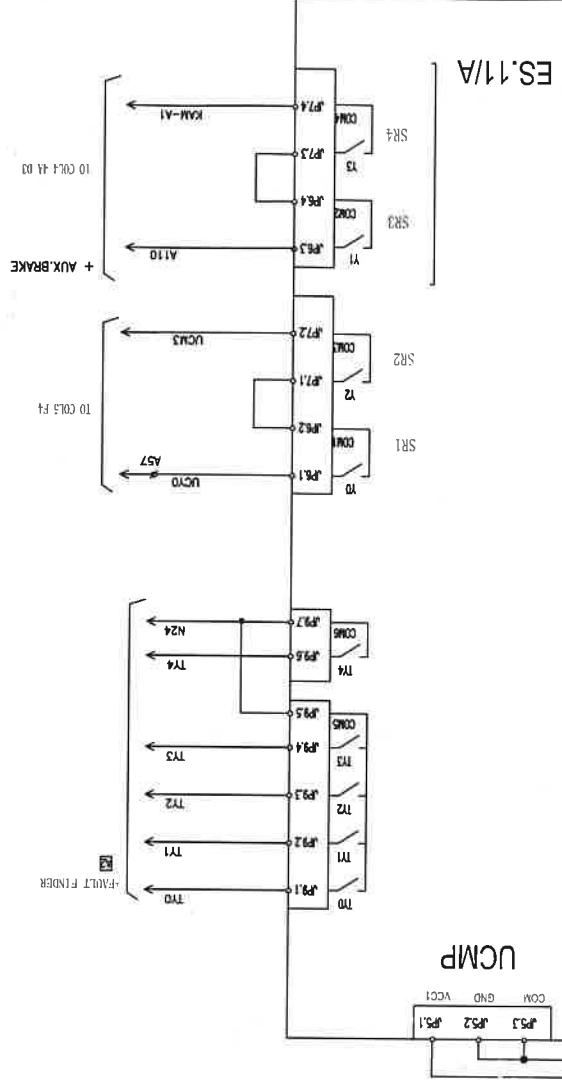
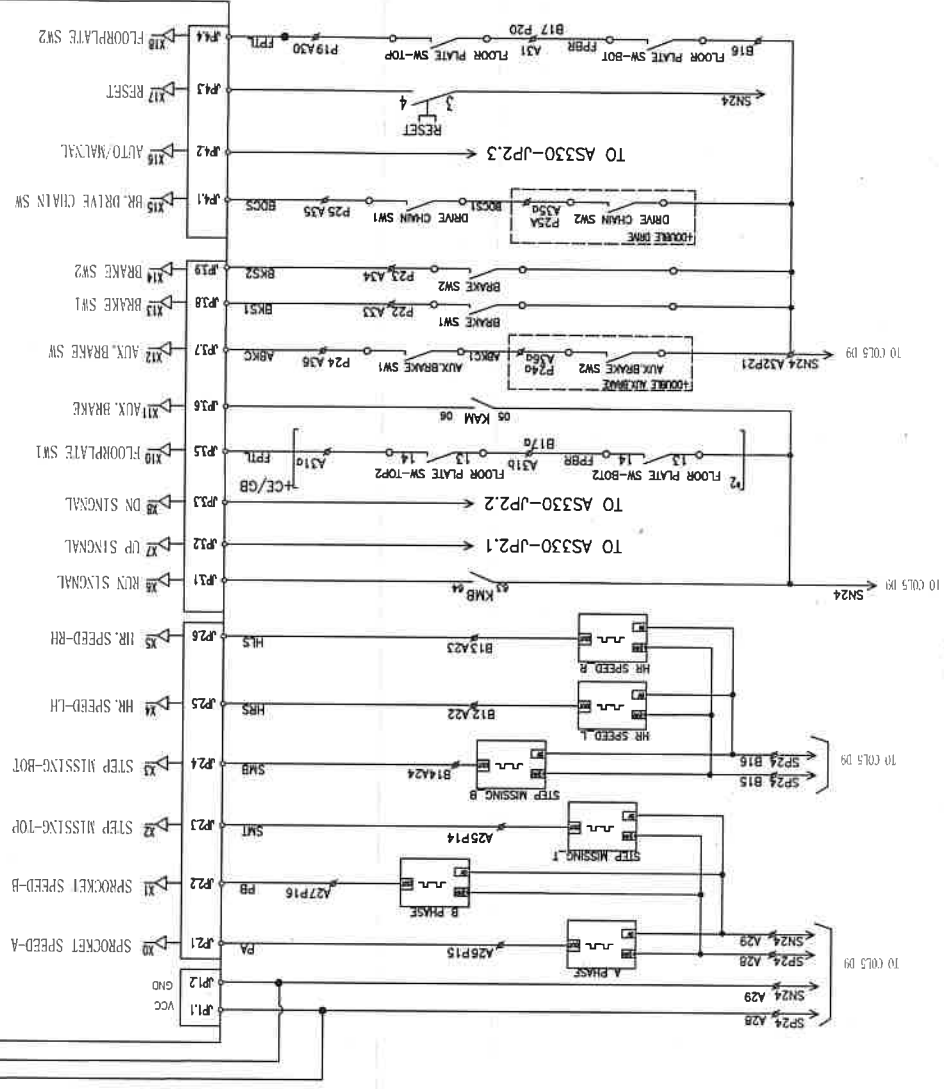
|   |         |         |         |
|---|---------|---------|---------|
| 사 | H.F.F   |         |         |
|   | DRWN BY | DSND BY | CHKD BY |
| 재 | 제 도     | 설 계     | 감 도     |
|   | APPD BY | 승 인     |         |



**HYUNDAI**  
ELEVATOR CO., LTD.

|           |         |
|-----------|---------|
| UNIT      | MM.     |
| SCALE     | 1" = 1' |
| 3RD ANGLE | 3 2 1   |

| APPAR. | MCU | TITLE                     | DATE |
|--------|-----|---------------------------|------|
|        |     | CONTROL CIRCUIT SCHEMATIC |      |



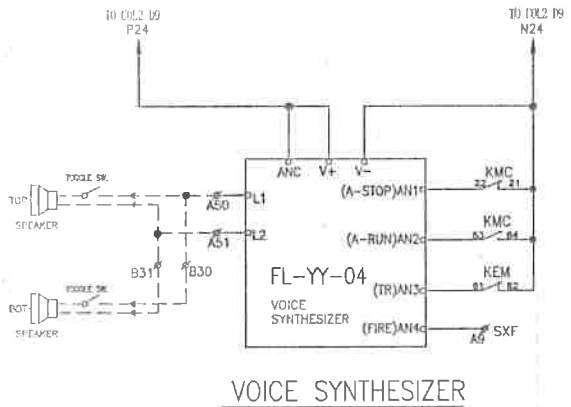
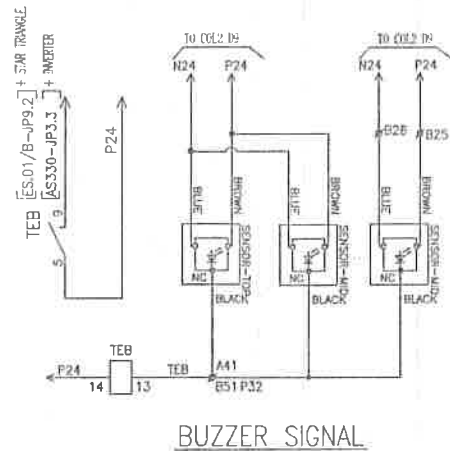
NOTE: THIS DRAWING APPLIED TO INVERTER.

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
해기없이 복사할 수 없음

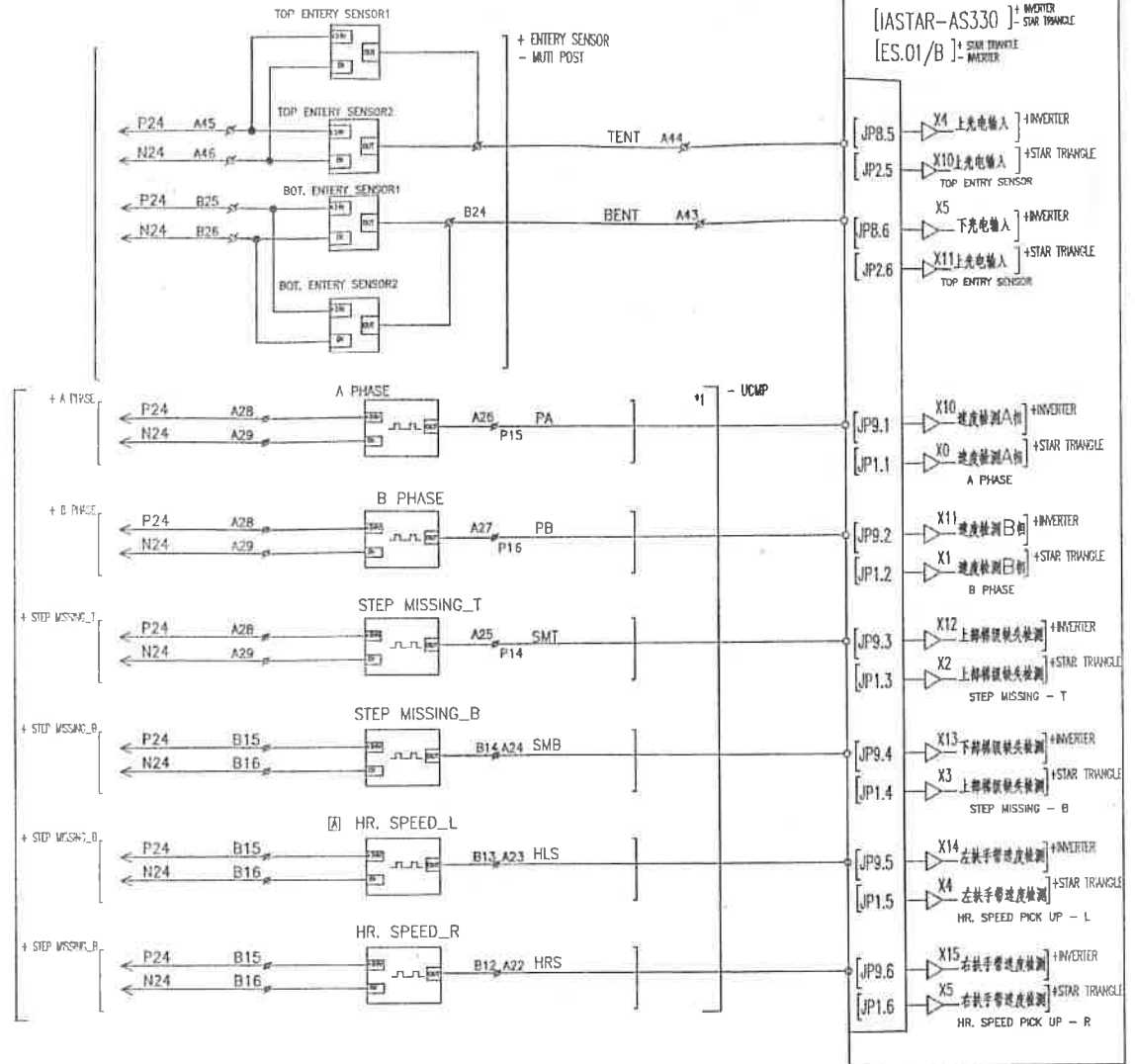
배 부 자

NOTE:  
\*1: WITHOUT UICMP, P24 IS CONNECTED TO A28 AND B15, N24 IS CONNECTED TO A29 AND B16.

CHANGE  
개정  
CHANGE HR. SPEED (E5)  
FL 2017.06.28



COL 6

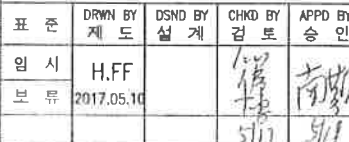


| VER. | DATE       | BY   | CHKD BY | APPD BY |
|------|------------|------|---------|---------|
| 1    | 2017.05.10 | H.FF |         |         |

| UNIT      | MM.   | APPR.   | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면  | 도면번호    |                           |

| UNIT      | MM.   | APPR.   | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면  | 도면번호    |                           |



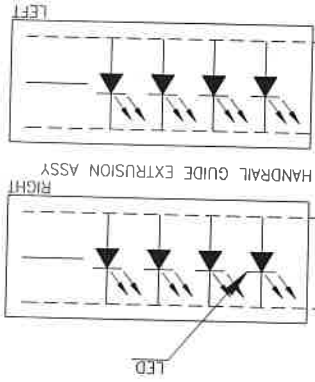
[illegible] $\frac{8}{19}$

|                                   |  |         |       |         |     |
|-----------------------------------|--|---------|-------|---------|-----|
|                                   |  |         | H. FF | 林杰      | 张兑圭 |
|                                   |  | DONG BY | 제 도   | CARD BY | 승 인 |
|                                   |  | DONG BY | 제 도   | CARD BY | 승 인 |
| <b>HYUNDAI ELEVATOR CO., LTD.</b> |  |         |       |         |     |

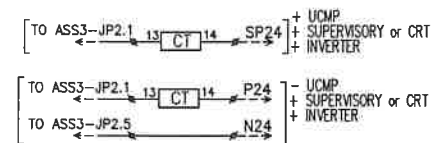
| APPAR | TITLE | 제목 |
|-------|-------|----|
|       |       |    |

|     |                           |  |
|-----|---------------------------|--|
| MCU | CONTROL CIRCUIT SCHEMATIC |  |
|-----|---------------------------|--|

11.5. 2019. 03. 05

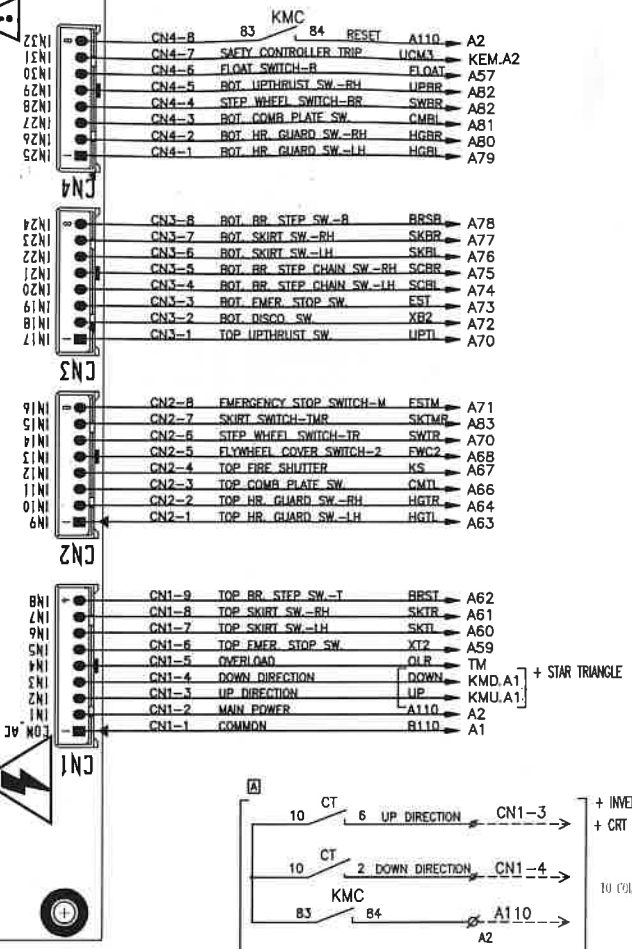


NOTE:

[illegible]

TO  
NIF-RMS

CRT SUPERVISORY BOARD



|                                                                                                                            |            |         |         |         |
|----------------------------------------------------------------------------------------------------------------------------|------------|---------|---------|---------|
|  <b>HYUNDAI</b><br>ELEVATOR CO., LTD. |            |         |         |         |
| 표                                                                                                                          | DRWN BY    | DSND BY | CHKD BY | APPD BY |
| 현                                                                                                                          | 제          | 도       | 검       | 승       |
|                                                                                                                            | 인          | 인       | 인       | 인       |
| 일                                                                                                                          | H.FF       |         |         |         |
| 시                                                                                                                          |            |         |         |         |
| 보                                                                                                                          | 2017.05.10 |         |         |         |
| 류                                                                                                                          |            |         |         |         |

|                    |       |                 |                           |
|--------------------|-------|-----------------|---------------------------|
| UNIT<br>단위         | MM.   | APPAR.          | MCU                       |
| SCALE<br>척도        | 1 / 1 | TITLE<br>제목     | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE<br>3 각 법 |       | REF. NO<br>참고도면 | DWG. NO<br>도면번호           |
|                    |       | CHE000057       | 10 / 19                   |

[illegible]

CRT COMMUNICATION BOARD

TO COL9 E5

POWER  
SUPPLY

POWER SUPPLY(DC5V)

| TOLERANCES—UNLESS OTHERWISE SPECIFIED |            |                                      |                                          |                           |               |
|---------------------------------------|------------|--------------------------------------|------------------------------------------|---------------------------|---------------|
| WORK                                  |            | TURN<br>PHOTO<br>OFFSET<br>HOLE STR. | MIS.<br>DE<br>COP.<br>PRECISION<br>DRAW. | SHAFT<br>WELD<br>GAS C/W. | CUT.<br>FORM. |
| SIZE                                  | TO<br>FIN. |                                      |                                          |                           |               |
| OVER                                  | TO<br>FIN. |                                      |                                          |                           |               |
| 1                                     | 4          | ± 0.1                                | ± 0.2                                    | ± 0.3                     | ± 0.3         |
| 4                                     | 16         | ± 0.2                                | ± 0.3                                    | ± 0.5                     | ± 0.7         |
| 16                                    | 63         | ± 0.3                                | ± 0.4                                    | ± 0.7                     | ± 0.9         |
| 63                                    | 250        | ± 0.4                                | ± 0.6                                    | ± 1.2                     | ± 1.3         |
| 250                                   | 500        | ± 0.5                                | ± 0.8                                    | ± 1.5                     | ± 1.6         |



|    |                |                |                |                |
|----|----------------|----------------|----------------|----------------|
| 표준 | DRWN BY<br>제 도 | DSND BY<br>설 계 | CHKD BY<br>검 토 | APPD BY<br>승 인 |
| 임시 | H.FF           |                |                |                |

|                    |       |                                       |    |
|--------------------|-------|---------------------------------------|----|
| UNIT<br>단위         | MM.   | APPAR. MCU                            |    |
| SCALE<br>척도        | 1 / 1 | TITLE<br>제목 CONTROL CIRCUIT SCHEMATIC |    |
| 3RD ANGLE<br>3 각 법 |       |                                       |    |
| REF. NO            |       | DWG. NO                               | 11 |



배 부 처



图纸确认意  
22 MAY 2017



**HYUNDAI**  
ELEVATOR CO., LTD.

|                    |             |                 |                                    |
|--------------------|-------------|-----------------|------------------------------------|
| UNIT<br>단위         | MM.         | APPAR.          | MCU                                |
| SCALE<br>척도        | 1<br>—<br>1 | TITLE<br>제목     | CONTROL CIRCUIT SCHEMATIC          |
| 3RD ANGLE<br>3 각 법 |             | REF. NO<br>참고도면 | DWG. NO<br>도면번호<br>CHE000057 12/19 |

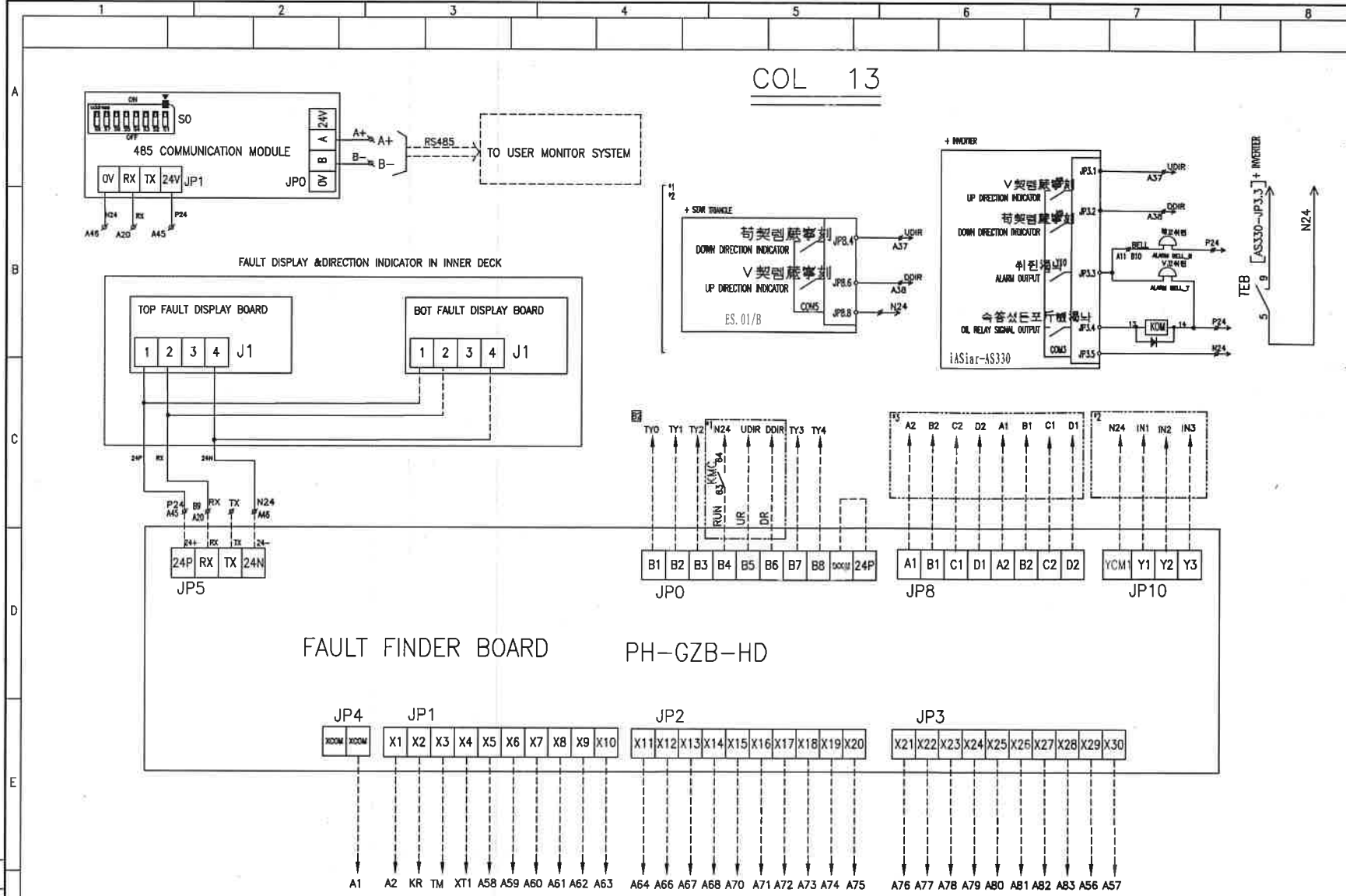


THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
허가없이 복사할 수 없음

COL 13

B1 FAULT DISPLAY CODE TABLE

| NO. | SAFETY SWITCH NAME                   | CODE  | REMARK                 |
|-----|--------------------------------------|-------|------------------------|
| 1   | PE PROTECTION                        | [011] |                        |
| 2   | MOTOR OVERLOAD ERROR                 | [012] |                        |
| 3   | MOTOR THERMISTOR ERROR               | [013] |                        |
| 4   | PHASE RELAY                          | [014] |                        |
| 5   | CONTROLLER STOP BUTTON               | [015] |                        |
| 6   | TOP STOP BUTTON                      | [016] |                        |
| 7   | TOP LEFT SHORT SW                    | [017] |                        |
| 8   | TOP RIGHT SHORT SW                   | [018] |                        |
| 9   | TOP BROKEN STEP SW                   | [019] |                        |
| 10  | TOP LEFT HIL GUARD SW                | [110] |                        |
| 11  | TOP RIGHT HIL GUARD SW               | [111] |                        |
| 12  | TOP COMB PLATE SW                    | [112] |                        |
| 13  | FIRE SW ERROR                        | [113] |                        |
| 14  | FLOOR COVER SW                       | [114] | OPTION                 |
| 15  | TOP UPTHURST SW                      | [115] | OPTION                 |
| 16  | MIDDLE STOP BUTTON                   | [116] | OPTION                 |
| 17  | BOT. DISCONNECT STOP BUTTON          | [117] |                        |
| 18  | BOT. STOP BUTTON                     | [118] |                        |
| 19  | BOT. LEFT BR. STEP CHAIN SW          | [119] |                        |
| 20  | BOT. RIGHT BR. STEP CHAIN SW         | [200] |                        |
| 21  | BOT. LEFT SHORT SW                   | [211] |                        |
| 22  | BOT. RIGHT SHORT SW-BR               | [212] |                        |
| 23  | BOT. BROKEN STEP SW                  | [213] |                        |
| 24  | BOT. LEFT HIL GUARD SW               | [214] |                        |
| 25  | BOT. RIGHT HIL GUARD SW              | [215] |                        |
| 26  | BOT COMB SW                          | [216] |                        |
| 27  | BOT. UPTHURST SW                     | [217] | OPTION                 |
| 28  | MIDDLE SHORT SW                      | [218] | OPTION                 |
| 29  | EARTHQUAKE SENSOR                    | [219] | OPTION                 |
| 30  | FLOAT SW                             | [300] | OPTION                 |
| NO. | UCIMP FAULT NAME                     | CODE  | REMARK                 |
| 31  | REAL-TIME CLOCK BREAKDOWN            | [311] | UCIMP INNER FAULT      |
| 32  | REDUNDANCY INSPECTION ERROR          | [312] | UCIMP INNER FAULT      |
| 33  | PARAMETER CRC ERROR                  | [313] | UCIMP INNER FAULT      |
| 34  | FAULT CRC ERROR                      | [314] | UCIMP INNER FAULT      |
| 35  | UNREASONABLE PARAMETER               | [315] | UNREASONABLY SET       |
| 36  | POWER VOLTAGE FAULT                  | [316] | VOLTAGE IS HIGH        |
| 37  | SAFETY RELAY FAULT                   | [317] | UCIMP INNER FAULT      |
| 38  | OVERSPEED MORE THAN 120%             | [318] | CHECK THE PARAMETERS   |
| 39  | OVERSPEED MORE THAN 140%             | [319] | AND THE ESCALATOR      |
| 40  | UNDEROPEED LOWER THAN 80%            | [400] | IS UNDEROPEEDING       |
| 41  | AB PHASE REVERSE CONNECTION          | [411] | CHECK RELAY SIGNAL     |
| 42  | UNUSUAL DROP OF SPEED                | [412] | THE "STOP" SIGNAL WHEN |
| 43  | DRIVE CHAIN DISCONNECTED             | [413] | TOP OF THE CABLE BEING |
| 44  | TRIPPING THE TROUBLE SHOOTING        | [414] | CHECK SIGNAL OF 312    |
| 45  | FAULT OF AUXILIARY BRAKE             | [415] | CHECK RELAY SIGNAL     |
| 46  | MISSING OF STEP (UP)                 | [416] | CHECK WHETHER IT IS    |
| 47  | MISSING OF STEP (DOWN)               | [417] | FUNCTIONING NORMAL     |
| 48  | MISSING OF FLOOR PLATE               | [418] | CHECK WHETHER IT IS    |
| 49  | FAULT OF FLOOR PLATE SWITCH          | [419] | AND THE SW OF ON       |
| 50  | WORK BRAKE CAN NOT OPENED            | [500] | CHECK WHETHER THE SW   |
| 51  | WORK BRAKE NOT CLOSED                | [501] | IS FUNCTIONING         |
| 52  | UNDEROPEED OF LEFT HANDRAIL          | [502] | CHECK WHETHER 513      |
| 53  | UNDEROPEED OF RIGHT HANDRAIL         | [503] | AND 514                |
| 54  | UNDEROPEED OF LEFT HANDRAIL          | [504] | IF HANDRAIL IS NORMAL  |
| 55  | OVERSPEED OF RIGHT HANDRAIL          | [505] | AND PHASE INVERTER AND |
| 56  | IF WORK BRAKE NOT CORRECTLY SET      | [506] | CHECK WHETHER IT IS    |
| 57  | SLIP DURING LEFT-STOP                | [507] | CHECK WHETHER IT IS    |
| 58  | THE 2 CH CHANNELS ARE NOT COORDINATE | [508] | IF 2 CH CHANNELS ARE   |
| 59  | LOSS OF AB PHASE                     | [509] | COORDINATE WITH A      |



NOTE:  
\*1: ADDED FOR FAULT DISPLAY WITH DIRECTION INDICATOR.  
\*2: ADDED FOR MULTI-POST ASP-A/S.  
\*3: ADDED FOR FAULT DISPLAY WITH MULTI-POST ASP-A/S.

ANG  
정관  
CODE TABLE (C-9)  
017.06.28  
CODE TABLE (A-9)  
—T74 (C-4)  
2017.10.20

TOLERANCES—UNLESS OTHERWISE SPECIFIED

| SIZE | TURN FROM OUT OF HOLE SUR. | MAX. DEPTH OF HOLE SUR. | SHAFT HOLD DIA. CHL. FORG. | CYL. FORG. |
|------|----------------------------|-------------------------|----------------------------|------------|
| 1    | ± 0.1                      | ± 0.2                   | ± 0.3                      | ± 0.2      |
| 2    | ± 0.2                      | ± 0.3                   | ± 0.5                      | ± 0.3      |
| 3    | ± 0.3                      | ± 0.4                   | ± 0.7                      | ± 0.5      |

**HYUNDAI ELEVATOR CO., LTD.**

표준 제도  
DRWN BY  
제도  
DSND BY  
실계  
CHKD BY  
검토  
APPD BY  
승인

UNIT  
단위  
MM.  
SCALE  
척도  
1/1

JRD ANGLE  
3 가 버

APPAR. MCU

TITLE  
제목  
CONTROL CIRCUIT SCHEMATIC

이제부터는  
복합형 △ 연호



**HYUNDAI**  
ELEVATOR CO., LTD.

|                    |     |
|--------------------|-----|
| UNIT<br>단위         | MM. |
| SCALE<br>척도        | 1   |
| 3RD ANGLE<br>3 각 법 |     |
| REF. NO<br>참고도면    |     |

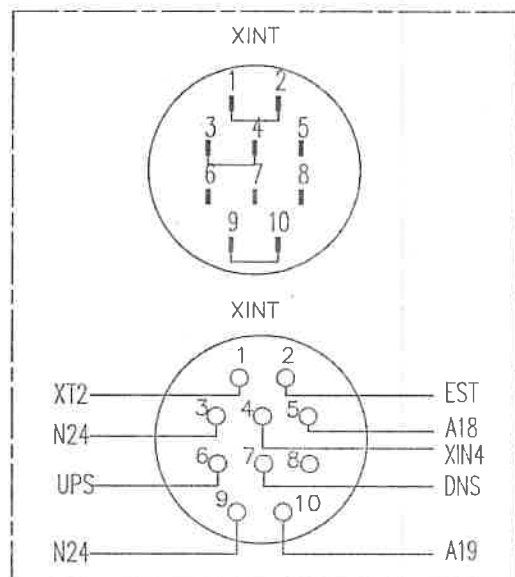
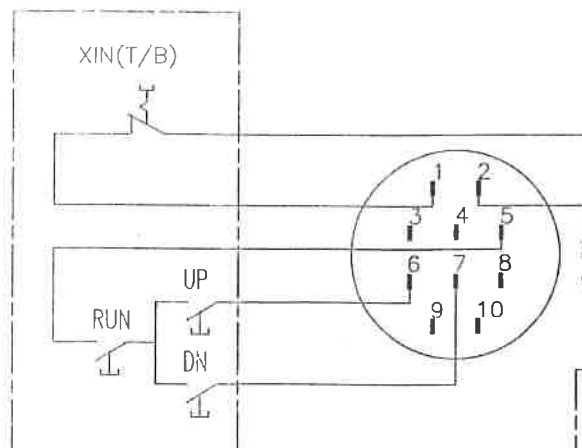
|   |         |                           |    |
|---|---------|---------------------------|----|
| 1 | APPAR.  | MCU                       |    |
|   | TITLE   | CONTROL CIRCUIT SCHEMATIC |    |
|   | 제목      |                           |    |
|   | DWG. NO | CHE000057                 | 14 |
|   | 도면번호    |                           | 1  |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
허가없이 복사할 수 없음

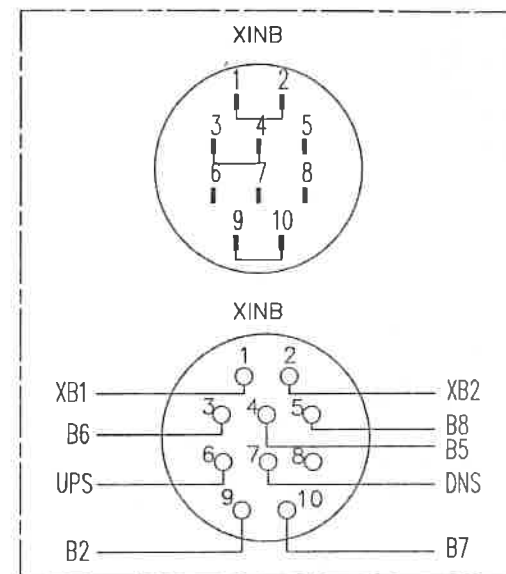
배 부 지

CHANGE  
개정란

检修手柄接线示意图  
INSPECTION HANDLE



上检修手柄插座接线示意图  
INSPECTION HANDLE SOCKET\_TOP



下检修手柄插座接线示意图  
INSPECTION HANDLE SOCKET\_BOT.

图纸确认章  
22 MAY 2017

| DIMENSIONS—IMPERIAL (OTHERWISE SPECIFIED) |      |         |                   |                   |                   |
|-------------------------------------------|------|---------|-------------------|-------------------|-------------------|
| NO.                                       | SIZE | TYPICAL | MAX. DIM. OF HOLE | MIN. DIM. OF HOLE | MAX. DIM. OF HOLE |
| 1                                         | 4    | 0.01    | 0.02              | 0.03              | 0.05              |
| 2                                         | 11   | 0.02    | 0.03              | 0.04              | 0.07              |
| 3                                         | 15   | 0.03    | 0.04              | 0.05              | 0.08              |
| 4                                         | 20   | 0.04    | 0.05              | 0.06              | 0.10              |
| 5                                         | 25   | 0.05    | 0.06              | 0.07              | 0.12              |

**HYUNDAI**  
ELEVATOR CO., LTD.

표준  
제도  
설계  
검토  
승인

DRWN BY  
H.F.F.

DSND BY

CHKD BY

APPD BY

| UNIT<br>단위         | MM. | APPAR.      | MCU                       |
|--------------------|-----|-------------|---------------------------|
| SCALE<br>척도        | 1/1 | TITLE<br>제목 | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE<br>3 각 법 |     | REF. NO     |                           |
| DWG. NO            |     |             |                           |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이것이 복사될 수 없음

배 부 지

CHANCE  
공정

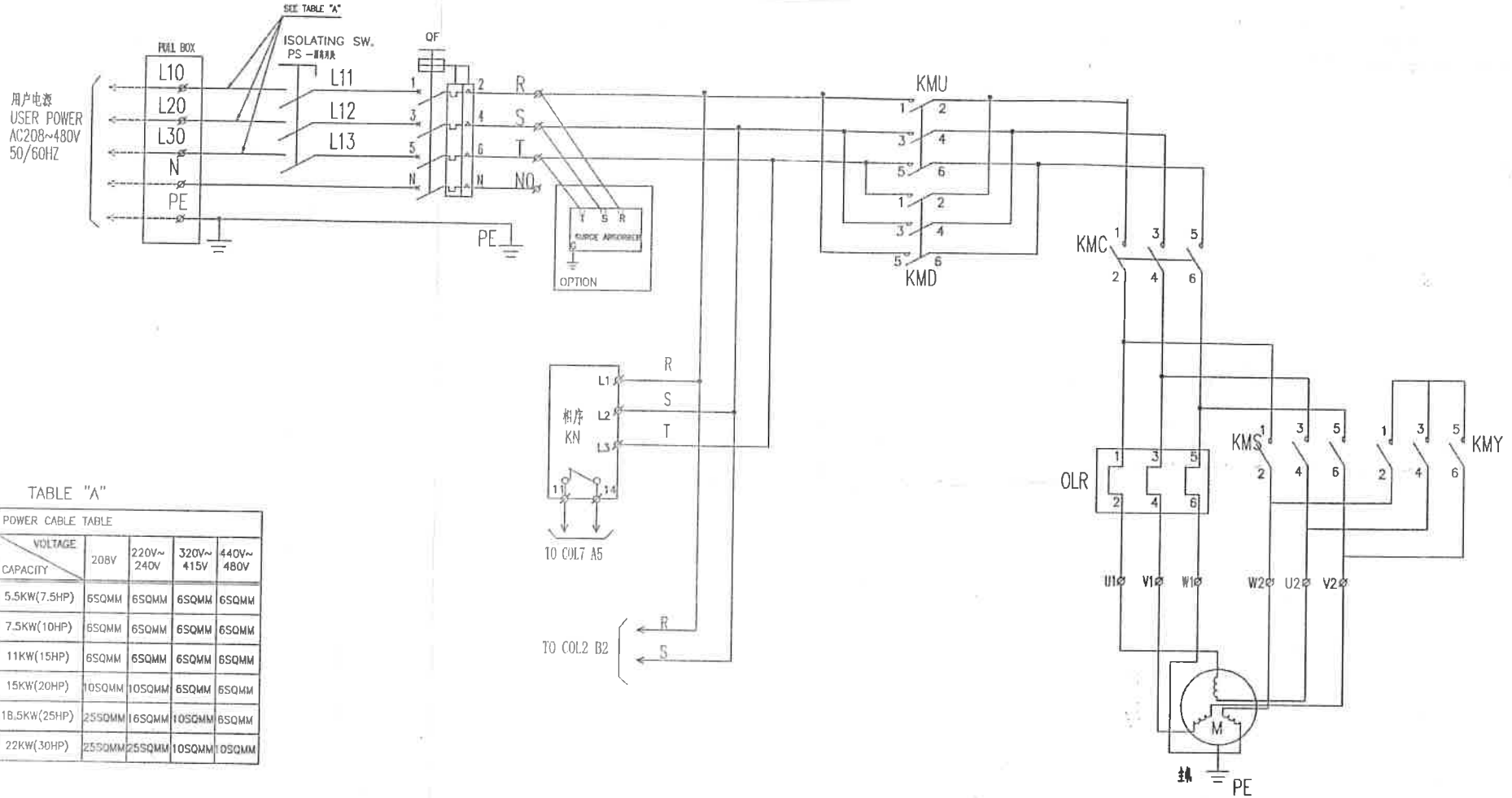


TABLE "A"

| CAPACITY     | VOLTAGE |           |           |           |
|--------------|---------|-----------|-----------|-----------|
|              | 208V    | 220V~240V | 320V~415V | 440V~480V |
| 5.5KW(7.5HP) | 6SQMM   | 6SQMM     | 6SQMM     | 6SQMM     |
| 7.5KW(10HP)  | 6SQMM   | 6SQMM     | 6SQMM     | 6SQMM     |
| 11KW(15HP)   | 6SQMM   | 6SQMM     | 6SQMM     | 6SQMM     |
| 15KW(20HP)   | 10SQMM  | 10SQMM    | 6SQMM     | 6SQMM     |
| 18.5KW(25HP) | 25SQMM  | 16SQMM    | 10SQMM    | 6SQMM     |
| 22KW(30HP)   | 25SQMM  | 25SQMM    | 10SQMM    | 10SQMM    |

图纸确认章  
22 MAY 2017

| WIRE | WIRE | WIRE | WIRE | WIRE |
|------|------|------|------|------|
| SIZE | TYPE | TYPE | TYPE | TYPE |
| 1    | 1.5  | 1.5  | 1.5  | 1.5  |
| 2    | 1.5  | 1.5  | 1.5  | 1.5  |
| 3    | 1.5  | 1.5  | 1.5  | 1.5  |
| 4    | 1.5  | 1.5  | 1.5  | 1.5  |
| 5    | 1.5  | 1.5  | 1.5  | 1.5  |
| 6    | 1.5  | 1.5  | 1.5  | 1.5  |
| 7    | 1.5  | 1.5  | 1.5  | 1.5  |
| 8    | 1.5  | 1.5  | 1.5  | 1.5  |
| 9    | 1.5  | 1.5  | 1.5  | 1.5  |
| 10   | 1.5  | 1.5  | 1.5  | 1.5  |
| 11   | 1.5  | 1.5  | 1.5  | 1.5  |
| 12   | 1.5  | 1.5  | 1.5  | 1.5  |
| 13   | 1.5  | 1.5  | 1.5  | 1.5  |
| 14   | 1.5  | 1.5  | 1.5  | 1.5  |
| 15   | 1.5  | 1.5  | 1.5  | 1.5  |
| 16   | 1.5  | 1.5  | 1.5  | 1.5  |
| 17   | 1.5  | 1.5  | 1.5  | 1.5  |
| 18   | 1.5  | 1.5  | 1.5  | 1.5  |
| 19   | 1.5  | 1.5  | 1.5  | 1.5  |
| 20   | 1.5  | 1.5  | 1.5  | 1.5  |
| 21   | 1.5  | 1.5  | 1.5  | 1.5  |
| 22   | 1.5  | 1.5  | 1.5  | 1.5  |
| 23   | 1.5  | 1.5  | 1.5  | 1.5  |
| 24   | 1.5  | 1.5  | 1.5  | 1.5  |
| 25   | 1.5  | 1.5  | 1.5  | 1.5  |
| 26   | 1.5  | 1.5  | 1.5  | 1.5  |
| 27   | 1.5  | 1.5  | 1.5  | 1.5  |
| 28   | 1.5  | 1.5  | 1.5  | 1.5  |
| 29   | 1.5  | 1.5  | 1.5  | 1.5  |
| 30   | 1.5  | 1.5  | 1.5  | 1.5  |
| 31   | 1.5  | 1.5  | 1.5  | 1.5  |
| 32   | 1.5  | 1.5  | 1.5  | 1.5  |
| 33   | 1.5  | 1.5  | 1.5  | 1.5  |
| 34   | 1.5  | 1.5  | 1.5  | 1.5  |
| 35   | 1.5  | 1.5  | 1.5  | 1.5  |
| 36   | 1.5  | 1.5  | 1.5  | 1.5  |
| 37   | 1.5  | 1.5  | 1.5  | 1.5  |
| 38   | 1.5  | 1.5  | 1.5  | 1.5  |
| 39   | 1.5  | 1.5  | 1.5  | 1.5  |
| 40   | 1.5  | 1.5  | 1.5  | 1.5  |
| 41   | 1.5  | 1.5  | 1.5  | 1.5  |
| 42   | 1.5  | 1.5  | 1.5  | 1.5  |
| 43   | 1.5  | 1.5  | 1.5  | 1.5  |
| 44   | 1.5  | 1.5  | 1.5  | 1.5  |
| 45   | 1.5  | 1.5  | 1.5  | 1.5  |
| 46   | 1.5  | 1.5  | 1.5  | 1.5  |
| 47   | 1.5  | 1.5  | 1.5  | 1.5  |
| 48   | 1.5  | 1.5  | 1.5  | 1.5  |
| 49   | 1.5  | 1.5  | 1.5  | 1.5  |
| 50   | 1.5  | 1.5  | 1.5  | 1.5  |
| 51   | 1.5  | 1.5  | 1.5  | 1.5  |
| 52   | 1.5  | 1.5  | 1.5  | 1.5  |
| 53   | 1.5  | 1.5  | 1.5  | 1.5  |
| 54   | 1.5  | 1.5  | 1.5  | 1.5  |
| 55   | 1.5  | 1.5  | 1.5  | 1.5  |
| 56   | 1.5  | 1.5  | 1.5  | 1.5  |
| 57   | 1.5  | 1.5  | 1.5  | 1.5  |
| 58   | 1.5  | 1.5  | 1.5  | 1.5  |
| 59   | 1.5  | 1.5  | 1.5  | 1.5  |
| 60   | 1.5  | 1.5  | 1.5  | 1.5  |
| 61   | 1.5  | 1.5  | 1.5  | 1.5  |
| 62   | 1.5  | 1.5  | 1.5  | 1.5  |
| 63   | 1.5  | 1.5  | 1.5  | 1.5  |
| 64   | 1.5  | 1.5  | 1.5  | 1.5  |
| 65   | 1.5  | 1.5  | 1.5  | 1.5  |
| 66   | 1.5  | 1.5  | 1.5  | 1.5  |
| 67   | 1.5  | 1.5  | 1.5  | 1.5  |
| 68   | 1.5  | 1.5  | 1.5  | 1.5  |
| 69   | 1.5  | 1.5  | 1.5  | 1.5  |
| 70   | 1.5  | 1.5  | 1.5  | 1.5  |
| 71   | 1.5  | 1.5  | 1.5  | 1.5  |
| 72   | 1.5  | 1.5  | 1.5  | 1.5  |
| 73   | 1.5  | 1.5  | 1.5  | 1.5  |
| 74   | 1.5  | 1.5  | 1.5  | 1.5  |
| 75   | 1.5  | 1.5  | 1.5  | 1.5  |
| 76   | 1.5  | 1.5  | 1.5  | 1.5  |
| 77   | 1.5  | 1.5  | 1.5  | 1.5  |
| 78   | 1.5  | 1.5  | 1.5  | 1.5  |
| 79   | 1.5  | 1.5  | 1.5  | 1.5  |
| 80   | 1.5  | 1.5  | 1.5  | 1.5  |
| 81   | 1.5  | 1.5  | 1.5  | 1.5  |
| 82   | 1.5  | 1.5  | 1.5  | 1.5  |
| 83   | 1.5  | 1.5  | 1.5  | 1.5  |
| 84   | 1.5  | 1.5  | 1.5  | 1.5  |
| 85   | 1.5  | 1.5  | 1.5  | 1.5  |
| 86   | 1.5  | 1.5  | 1.5  | 1.5  |
| 87   | 1.5  | 1.5  | 1.5  | 1.5  |
| 88   | 1.5  | 1.5  | 1.5  | 1.5  |
| 89   | 1.5  | 1.5  | 1.5  | 1.5  |
| 90   | 1.5  | 1.5  | 1.5  | 1.5  |
| 91   | 1.5  | 1.5  | 1.5  | 1.5  |
| 92   | 1.5  | 1.5  | 1.5  | 1.5  |
| 93   | 1.5  | 1.5  | 1.5  | 1.5  |
| 94   | 1.5  | 1.5  | 1.5  | 1.5  |
| 95   | 1.5  | 1.5  | 1.5  | 1.5  |
| 96   | 1.5  | 1.5  | 1.5  | 1.5  |
| 97   | 1.5  | 1.5  | 1.5  | 1.5  |
| 98   | 1.5  | 1.5  | 1.5  | 1.5  |
| 99   | 1.5  | 1.5  | 1.5  | 1.5  |
| 100  | 1.5  | 1.5  | 1.5  | 1.5  |

HYUNDAI  
ELEVATOR CO., LTD.

|    |            |         |         |         |
|----|------------|---------|---------|---------|
| 표준 | DRWN BY    | DSMD BY | CHKD BY | APPD BY |
| 제도 | H.F.F      |         |         |         |
| 실개 | 2017.05.10 |         |         |         |
| 검토 |            |         |         |         |
| 승인 |            |         |         |         |

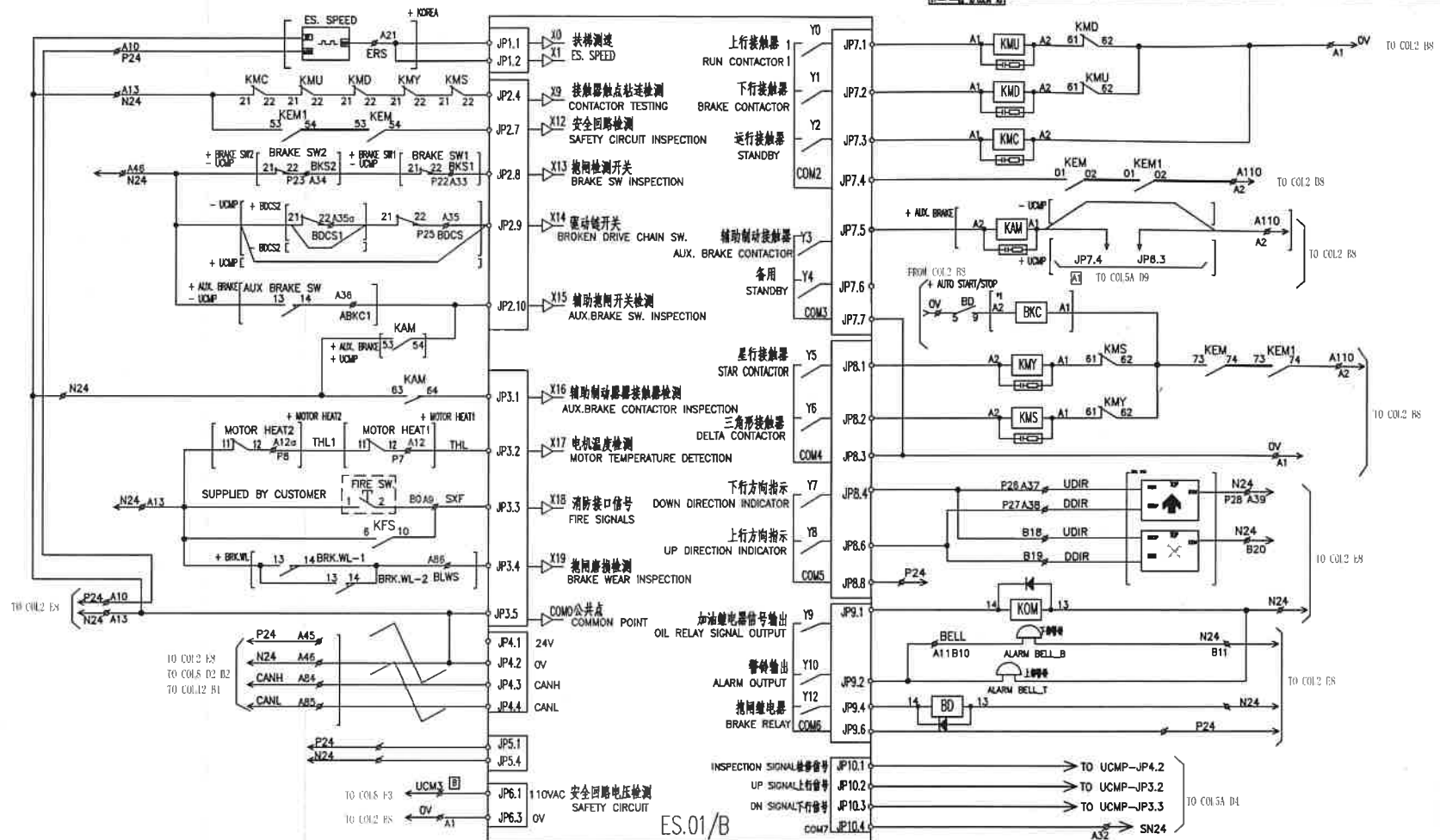
|           |     |
|-----------|-----|
| UNIT      | MM. |
| 단위        | mm. |
| SCALE     | 1/1 |
| 척도        | 1/1 |
| 3RD ANGLE |     |
| 3 각 법     |     |
| REF. NO   |     |
| 참고도면      |     |

|         |                           |
|---------|---------------------------|
| APPAR.  | MCU                       |
| TITLE   | CONTROL CIRCUIT SCHEMATIC |
| DWG. NO | CHE000057                 |
| 도면번호    |                           |

배 부 치

COL 4A

NOTE: THIS DRAWING APPLIED TO STAR TRIANGLE.



\*1:USED IF PROVIDED WITH MULTI-POST OR AUTO START/STOP(EXCEPT CHINA).

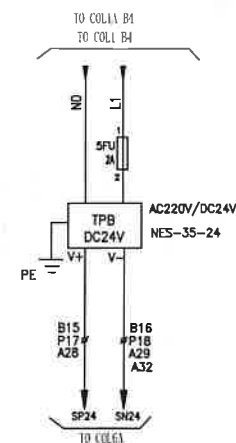
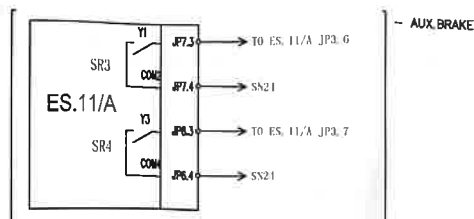
| WORK |            | TURN<br>PROF.<br>DEPT.<br>HOLE STR. | MOLD.<br>OF CAST.<br>PROLIFER<br>DRWG. | SHAFT<br>WELD<br>GWS CTS. | SPCL.<br>FORM. |
|------|------------|-------------------------------------|----------------------------------------|---------------------------|----------------|
| SIZE | TO<br>FIN. |                                     |                                        |                           |                |
| CHAM | TO<br>FIN. |                                     |                                        |                           |                |
| 1    | 4          | ± 0.1                               | ± 0.2                                  | ± 0.3                     | ± 0.5          |
| 4    | 16         | ± 0.2                               | ± 0.3                                  | ± 0.5                     | ± 0.7          |
| 16   | 63         | ± 0.3                               | ± 0.6                                  | ± 0.7                     | ± 0.9          |
| 63   | 250        | ± 0.4                               | ± 0.8                                  | ± 1.2                     | ± 1.5          |
| 250  | 1000       | ± 0.6                               | ± 0.8                                  | ± 1.5                     | ± 1.8          |



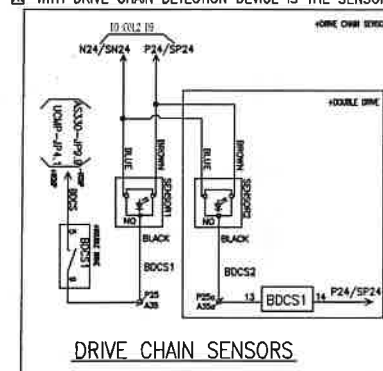
|                    |       |
|--------------------|-------|
| UNIT<br>단위         | MM.   |
| SCALE<br>척도        | 1 / 1 |
| 3RD ANGLE<br>3 각 법 |       |
| REF. NO            |       |

|         |                           |
|---------|---------------------------|
| APPAR.  | MCU                       |
| TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 제목      |                           |
| DWG. NO | 17                        |

NOTE: THIS DRAWING APPLIED TO STAR TRIANGLE.



**A1** \*WITH DRIVE CHAIN DETECTION DEVICE IS THE SENSORS.



**HYUNDAI**  
ELEVATOR CO., LTD.

|                    |       |
|--------------------|-------|
| UNIT<br>단위         | MM.   |
| SCALE<br>척도        | 1 / 1 |
| 3RD ANGLE<br>3 각 법 |       |
| REF. NO<br>참고도면    |       |

|         |                           |    |
|---------|---------------------------|----|
| APPAR.  | MCU                       |    |
| TITLE   | CONTROL CIRCUIT SCHEMATIC |    |
| 제목      |                           |    |
| DWG. NO | CH50000057                | 18 |

18/19

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
허가없이 복사할 수 없음

CHANGE 개정 라

ADDED DRIVE CHAIN SENSOR(E,6)  
ADDED TYO---TY4 (B,7)  
H.FF 2017.10.20

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
해결되지 않음

# COL 16

## INDEX & LABEL LIST

61

| ITEM                       | SYMBOL | REMARK       | ITEM                  | SYMBOL | REMARK       |
|----------------------------|--------|--------------|-----------------------|--------|--------------|
| 절연 스위치<br>ISOLATING SWITCH |        |              | 배전 코일<br>COIL         |        |              |
| 회로 차단기<br>CIRCUIT BREAKER  |        |              | 접촉기<br>CONNECTOR      |        |              |
| 접촉기<br>CONNECTOR           |        |              | 키 스위치<br>KEY SW.      |        |              |
| A CONTACT                  |        | MAIN CONTACT | 버튼<br>BUTTON          |        |              |
| FUSE                       |        |              | 입출 신호<br>INPUT SIGNAL |        |              |
| TERMINAL BLOCK             |        |              | 센서<br>SENSOR          |        |              |
| MOTOR ROOM HEAD PUSHBUTTON |        |              | B CONTACT             |        | AUX. CONTACT |
| STOP BUTTON                |        |              | C CONTACT             |        | AUX. CONTACT |
| LIMIT SWITCH               |        |              | D CONTACT             |        |              |
| RELAY & CONTACTOR COIL     |        |              | 회전선<br>TURNED LINE    |        |              |
|                            |        |              | FAN                   |        |              |
|                            |        |              | 수용기<br>RECEPTACLE     |        |              |
|                            |        |              | 지중<br>EARTH           |        |              |
|                            |        |              | 벨<br>BELL             |        |              |
|                            |        |              | 마그네틱 링<br>MAGNET RING |        |              |

| LABEL      | COL NO.    | LABEL DESCRIPTION        |
|------------|------------|--------------------------|
| OF         | COL1/COL1A | AIR BREAK SWITCH         |
| KN         | COL1/COL1A | PHASE DETECTOR           |
| KR         | COL1A      | OVERLOAD PROTECTOR       |
| RB         | COL1       | INVERTER BRAKE RESISTOR  |
| TCO        | COL2       | TRANSFORMER              |
| PS         | COL2       | ISLATING SWITCH          |
| CPS/TPB    | COL2/COL5  | POWER SWITCH             |
| FU/FA      | COL2       | FUSE                     |
| BK         | COL2       | BRAKE                    |
| ABK        | COL2       | AUX.BRAKE                |
| OIL        | COL2       | OIL                      |
| XINT       | COL3       | TOP INSPECTION SOCKET    |
| XINB       | COL3       | BOT.INSPECTION SOCKET    |
| KF/KFS/SHR | COL3/COL2  | RELAY                    |
| STU/STD    | COL3       | TOP/BOT.KEY SWITCH       |
| SXF        | COL4/COL4A | FIRE                     |
| THL        | COL4/COL4A | MOTOR THERMISTOR         |
| ERS        | COL4/COL4A | SPEED SENSOR             |
| KMU/KMD    | COL4A      | UP/DOWN CONTACTOR        |
| KEM        | COL7       | SAFETY CIRCUIT CONTACTOR |
| BELL       | COL4/COL4A | BELL                     |
| KAM        | COL4/COL4A | AUX.BRAKE CONTACTOR      |
| KMB        | COL4       | BRAKE CONTACTOR          |
| BKC        | COL4A      | BRAKE CONTACTOR          |
| BD         | COL4A      | BRAKE RELAY              |
| KOM        | COL4/COL4A | OIL RELAY                |
| UCMP       | COL5/COL5A | SAFETY MONITORING BOARD  |
| PA/PB      | COL5/COL5A | A/B PHASE                |
| SMT/SMB    | COL5/COL5A | TOP/BOT.STEP MISSING     |
| HRS/HLS    | COL5/COL5A | HANDRAIL SPEED SENSOR    |
| RESET      | COL5/COL5A | RESET BUTTON             |

| LABEL          | COL NO.    | LABEL DESCRIPTION       |
|----------------|------------|-------------------------|
| ABKC           | COL5/COL5A | AUX.BRAKE SWITCH        |
| BKS            | COL5/COL5A | BRAKE SWITCH            |
| FPBR/FPTL      | COL5/COL5A | FLOOR PLATE SWITCH      |
| FL-YY-04       | COL6       | VOICE SYNTHESIZER       |
| UDIR/DDIR      | COL4/COL4A | DIRECTION INDICATOR     |
| TEB            | COL6       | BUZZER SIGNAL SENSOR    |
| TENT/BENT      | COL6       | TOP/BOT.ENTRY SENSOR    |
| ASP            | COL12      | MULTI-POST              |
| TLKS/BLKS      | COL9       | LIGHTING KEY SWITCH     |
| KL1/KL2        | COL9       | LIGHTING RELAY          |
| LS             | COL9       | AIR SWITCH              |
| XS1/XS2        | COL9       | TOP/BOT. SOCKET         |
| HUOR METER     | COL9       | HUOR METER              |
| COUNTER        | COL9       | COUNTER                 |
| PH-GZB-HD      | COL13      | FAULT FINDER BOARD      |
| FDB            | COL13      | FAULT DISPLAY BOARD     |
| NIF-RMS EB     | COL10      | CRT SUPERVISORY BOARD   |
| NIF-RMS        | COL11      | CRT COMMUNICATION BOARD |
| KTO            | COL14      | TEMPERATURE CONTROLLER  |
| HEATER         | COL14      | HEATER                  |
| HR.HEATER      | COL14      | HANDRAIL HEATER         |
| HFN            | COL14      | BOX FAN                 |
| iASiar-AS330   | COL1/COL4  | INVERTER                |
| ES.01/B        | COL4A      | MAIN BAORD              |
| SURGE ABSORBER | COL1/COL1A | SURGE ABSORBER          |
| UCMP           | COL5/COL5A | SAFETY MONITOR BAORD    |
| ES.02/F        | COL8       | SAFETY MONITOR BAORD    |

CHANGE  
개정

图纸确认章  
22 MAY 2017

HYUNDAI  
ELEVATOR CO., LTD

표준  
제도  
설계  
검토  
승인  
H.FF  
2017.05.10

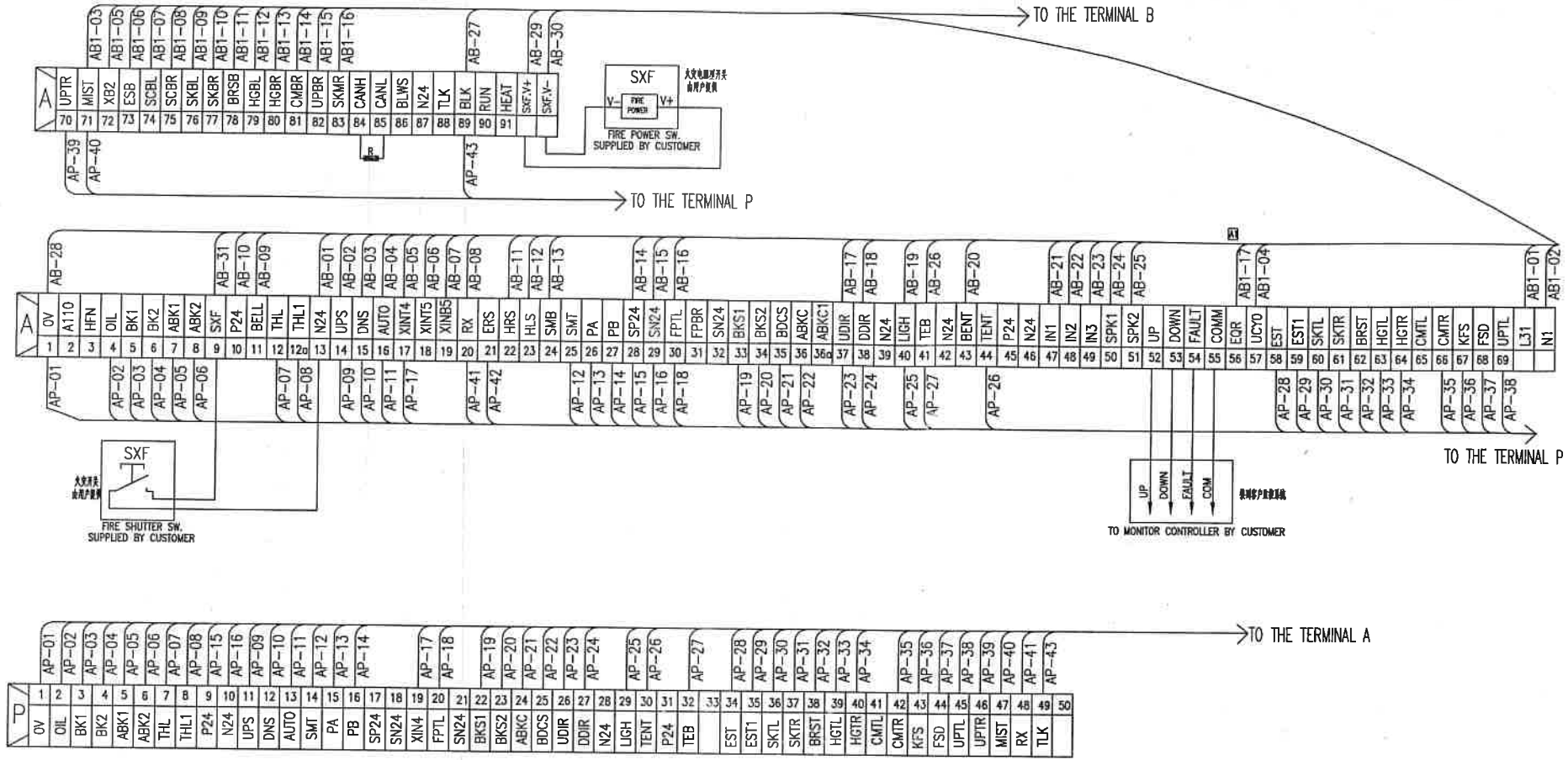
UNIT  
단위  
SCALE  
척도  
3RD ANGLE  
3 각 법  
REF. NO.  
DWC. NO.

APPAR. MCU  
TITLE CONTROL CIRCUIT SCHEMATIC  
DWC. NO. 015000057



THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이/그림이 복사될 수 없음

COL 1



Expected delivery time of the user

- NOTE:
1. THE TERMINALS A, P, B RESPECTIVELY ARE: THE CONTROLLER BOX, THE PULL BOX AND THE DISCONNECT BOX.
  2. THE TERMINAL P IS USED FOR THE SPECIAL STATUS.
  3. THE OPTIONAL FUNCTION TERMINAL CAN NOT INSTALL WITHOUT THIS OPTIONAL FUNCTION.
  4. LEISURE LOW SPEED TIME: T+60s. LEISURE STOP TIME: 8s. T IS EXPECTED DELIVERY TIME OF THE USER.

|                         |  |                   |  |
|-------------------------|--|-------------------|--|
| CHANGE<br>개 정 란         |  | H.F.F 2017.10.20  |  |
| CHANGE A56 TO EOR (C.B) |  | ADDED NOTE* (7.7) |  |
| H.F.F 2017.10.20        |  |                   |  |

|                       |            |                                       |                                             |
|-----------------------|------------|---------------------------------------|---------------------------------------------|
| TELEWORKS-ORDER SHEET |            | HYUNDAI<br>ELEVATOR CO., LTD.         |                                             |
| NO.                   | DATE       | TURN<br>PUMP'S<br>SET OF<br>ELE. SET. | M.O.D.<br>THE<br>ELE. SET<br>PUMP'S<br>SET. |
| 1                     | 2017.05.10 | 2.01                                  | 2.02                                        |
| 2                     | 2017.05.10 | 2.03                                  | 2.04                                        |
| 3                     | 2017.05.10 | 2.05                                  | 2.06                                        |
| 4                     | 2017.05.10 | 2.07                                  | 2.08                                        |
| 5                     | 2017.05.10 | 2.09                                  | 2.10                                        |
| 6                     | 2017.05.10 | 2.11                                  | 2.12                                        |
| 7                     | 2017.05.10 | 2.13                                  | 2.14                                        |
| 8                     | 2017.05.10 | 2.15                                  | 2.16                                        |
| 9                     | 2017.05.10 | 2.17                                  | 2.18                                        |
| 10                    | 2017.05.10 | 2.19                                  | 2.20                                        |
| 11                    | 2017.05.10 | 2.21                                  | 2.22                                        |
| 12                    | 2017.05.10 | 2.23                                  | 2.24                                        |
| 13                    | 2017.05.10 | 2.25                                  | 2.26                                        |
| 14                    | 2017.05.10 | 2.27                                  | 2.28                                        |
| 15                    | 2017.05.10 | 2.29                                  | 2.30                                        |
| 16                    | 2017.05.10 | 2.31                                  | 2.32                                        |
| 17                    | 2017.05.10 | 2.33                                  | 2.34                                        |
| 18                    | 2017.05.10 | 2.35                                  | 2.36                                        |
| 19                    | 2017.05.10 | 2.37                                  | 2.38                                        |
| 20                    | 2017.05.10 | 2.39                                  | 2.40                                        |
| 21                    | 2017.05.10 | 2.41                                  | 2.42                                        |
| 22                    | 2017.05.10 | 2.43                                  | 2.44                                        |
| 23                    | 2017.05.10 | 2.45                                  | 2.46                                        |
| 24                    | 2017.05.10 | 2.47                                  | 2.48                                        |
| 25                    | 2017.05.10 | 2.49                                  | 2.50                                        |
| 26                    | 2017.05.10 | 2.51                                  | 2.52                                        |
| 27                    | 2017.05.10 | 2.53                                  | 2.54                                        |
| 28                    | 2017.05.10 | 2.55                                  | 2.56                                        |
| 29                    | 2017.05.10 | 2.57                                  | 2.58                                        |
| 30                    | 2017.05.10 | 2.59                                  | 2.60                                        |
| 31                    | 2017.05.10 | 2.61                                  | 2.62                                        |
| 32                    | 2017.05.10 | 2.63                                  | 2.64                                        |
| 33                    | 2017.05.10 | 2.65                                  | 2.66                                        |
| 34                    | 2017.05.10 | 2.67                                  | 2.68                                        |
| 35                    | 2017.05.10 | 2.69                                  | 2.70                                        |
| 36                    | 2017.05.10 | 2.71                                  | 2.72                                        |
| 37                    | 2017.05.10 | 2.73                                  | 2.74                                        |
| 38                    | 2017.05.10 | 2.75                                  | 2.76                                        |
| 39                    | 2017.05.10 | 2.77                                  | 2.78                                        |
| 40                    | 2017.05.10 | 2.79                                  | 2.80                                        |
| 41                    | 2017.05.10 | 2.81                                  | 2.82                                        |
| 42                    | 2017.05.10 | 2.83                                  | 2.84                                        |
| 43                    | 2017.05.10 | 2.85                                  | 2.86                                        |
| 44                    | 2017.05.10 | 2.87                                  | 2.88                                        |
| 45                    | 2017.05.10 | 2.89                                  | 2.90                                        |
| 46                    | 2017.05.10 | 2.91                                  | 2.92                                        |
| 47                    | 2017.05.10 | 2.93                                  | 2.94                                        |
| 48                    | 2017.05.10 | 2.95                                  | 2.96                                        |
| 49                    | 2017.05.10 | 2.97                                  | 2.98                                        |
| 50                    | 2017.05.10 | 2.99                                  | 3.00                                        |

|                    |     |                 |                      |
|--------------------|-----|-----------------|----------------------|
| UNIT<br>단위         | MM. | APPR.           | MCU                  |
| SCALE<br>척도        | 1/1 | TITLE<br>제목     | CONTROL TRUSS WIRING |
| 3RD ANGLE<br>3 각 법 |     | DWG. NO<br>도면번호 | CHE000058            |
| REF. NO<br>참고도면    |     |                 | 1/2                  |







THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
허가없이 복사할 수 없음

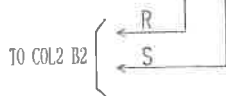
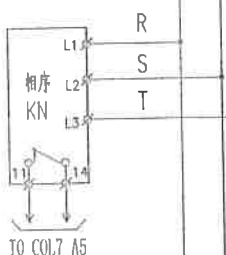
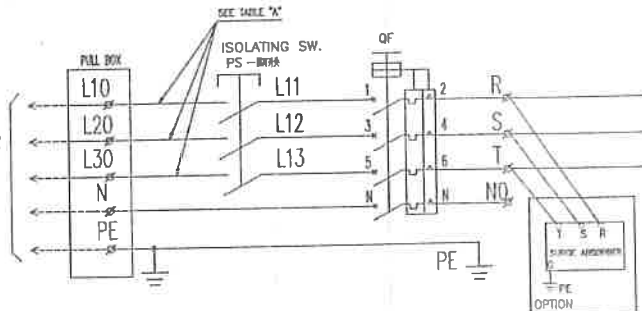
배 부 지

CHANGE  
개정판

用户电源  
USER POWER  
AC380~440V  
50/60HZ

TABLE "A"

| POWER CABLE TABLE   |        |        |  |
|---------------------|--------|--------|--|
| VOLTAGE<br>CAPACITY | 380V~  | 440V~  |  |
|                     | 415V   | 460V   |  |
| 5.5KW(7.5HP)        | 6SQMM  | 6SQMM  |  |
| 7.5KW(10HP)         | 6SQMM  | 6SQMM  |  |
| 11KW(15HP)          | 6SQMM  | 6SQMM  |  |
| 15KW(20HP)          | 10SQMM | 6SQMM  |  |
| 18.5KW(25HP)        | 10SQMM | 10SQMM |  |
| 22KW(30HP)          | 16SQMM | 10SQMM |  |
| 30KW(40HP)          | 16SQMM | 16SQMM |  |
| 37KW(50HP)          | 25SQMM | 16SQMM |  |



COL 1

NOTE: THIS DRAWING APPLIED TO INVERTER.

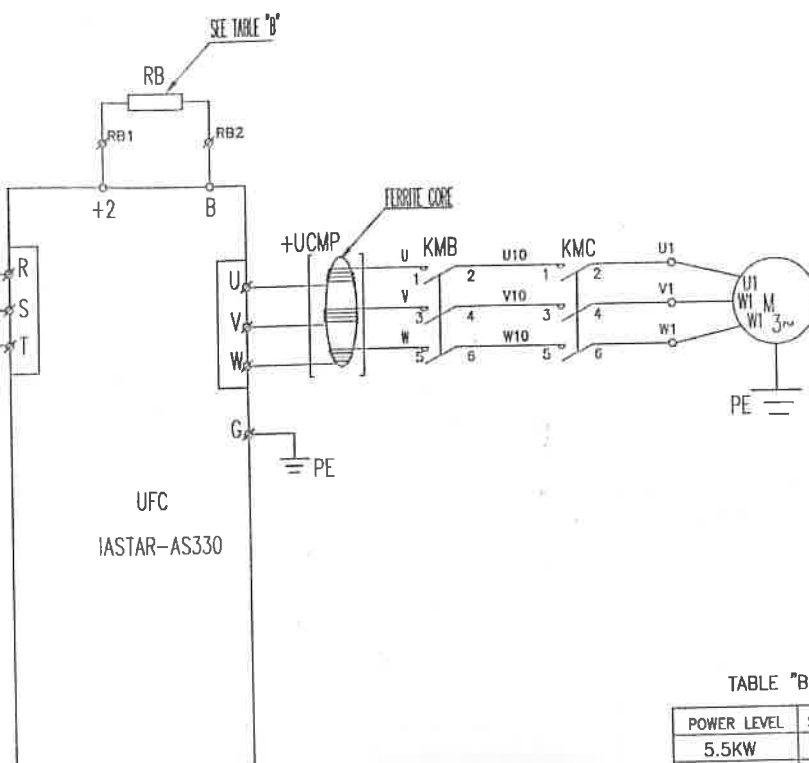


TABLE "B"

| POWER LEVEL | SPECIFICATIONS |
|-------------|----------------|
| 5.5KW       | 2000W, 70A     |
| 7.5KW       | 2000W, 64A     |
| 11KW        | 2000W, 40A     |
| 15KW        | 4000W, 36A     |
| 18.5KW      | 6000W, 24A     |
| 22KW        | 6000W, 20A     |
| 30KW        | 8000W, 15A     |

NOTE:  
\*1: ADDED FOR DOUBLE DRIVE.

图纸确认章  
22 MAY 2017

HYUNDAI  
ELEVATOR CO., LTD.

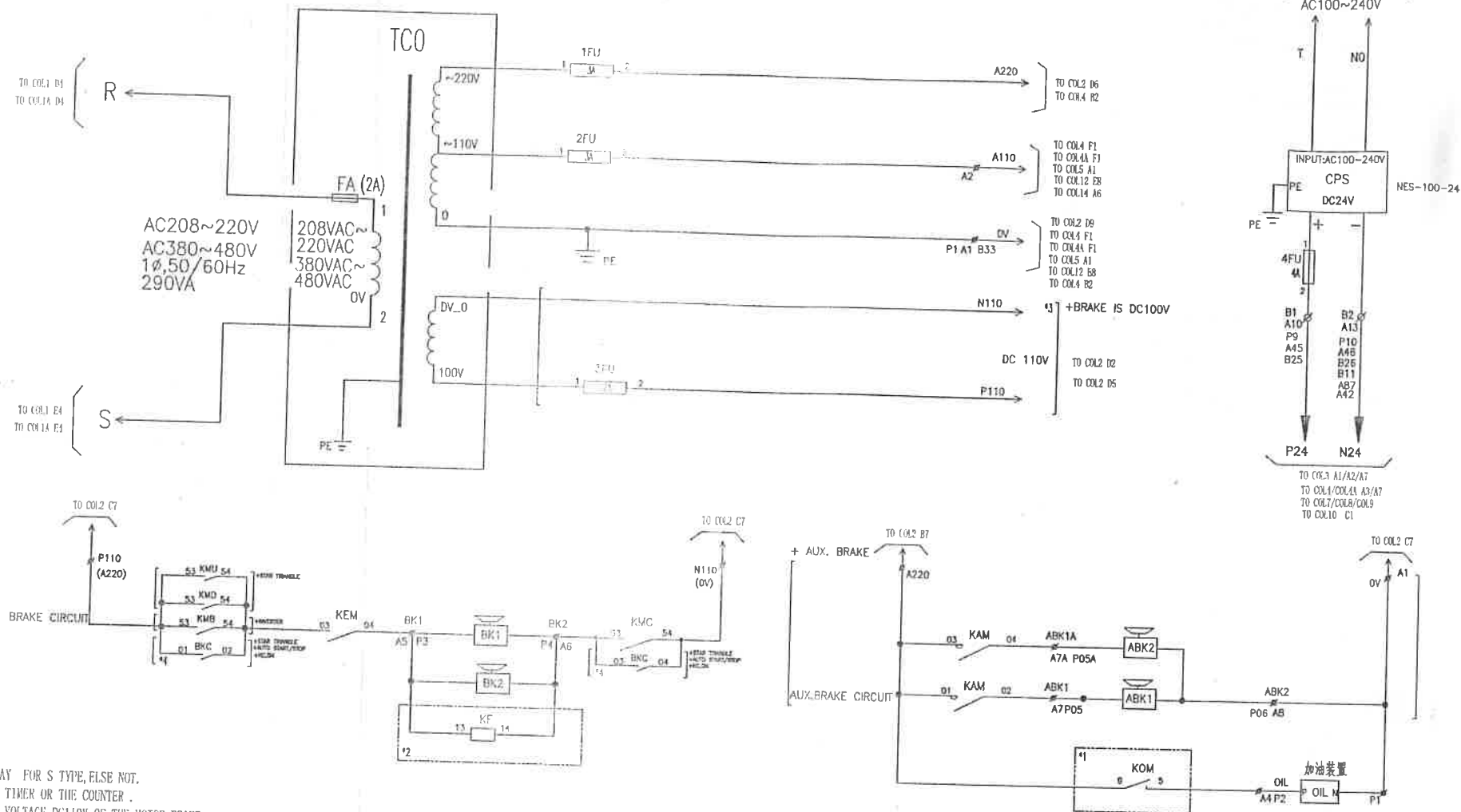
| 표준 | DRWN BY<br>제 도 | DSND BY<br>설 계 | CHKD BY<br>검 토 | APPD BY<br>승 인 |
|----|----------------|----------------|----------------|----------------|
| 임시 | H.FF           |                | 100<br>100     | 100            |
| 보류 | 2017.05.10     |                |                |                |

| UNIT<br>단위         | MM. |
|--------------------|-----|
| SCALE<br>척도        | 1/1 |
| 3RD ANGLE<br>3 각 법 |     |
| REF. NO<br>참고도면    |     |

| APPAR.          | MCU                       |
|-----------------|---------------------------|
| TITLE<br>제목     | CONTROL CIRCUIT SCHEMATIC |
| DWG. NO<br>도면번호 | CHE000057                 |

THIS DRAWING IS PROPRIETARY TO HELCO.  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO.  
각기장이 복사할 수 없음

COL 2



NOTE:

- \*1: ADDED KOM RELAY FOR S TYPE, ELSE NOT.
- \*2: ADDED FOR THE TIMER OR THE COUNTER.
- \*3: ADDED FOR THE VOLTAGE DC110V OF THE MOTOR BRAKE.
- \*4: USED IF PROVIDED WITH MULTI-POST OR AUTO START/STOP (EXCEPT CHINA)

| NO. | ITEM | UNIT | QTY | REMARK |
|-----|------|------|-----|--------|
| 1   | 1FU  | 1    | 1   |        |
| 2   | 2FU  | 1    | 1   |        |
| 3   | 3FU  | 1    | 1   |        |
| 4   | 4FU  | 1    | 1   |        |

HYUNDAI  
ELEVATOR CO., LTD

표준  
제도  
설계  
검토  
승인  
H.F.F.  
2017.05.10

| UNIT      | MM.  | APPR.   | MCU                       |
|-----------|------|---------|---------------------------|
| SCALE     | 1/1  | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면 | 도면번호    | 2/19                      |

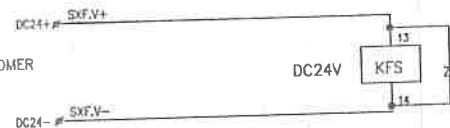
THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
허가없이 복사할 수 없음

배 부 적

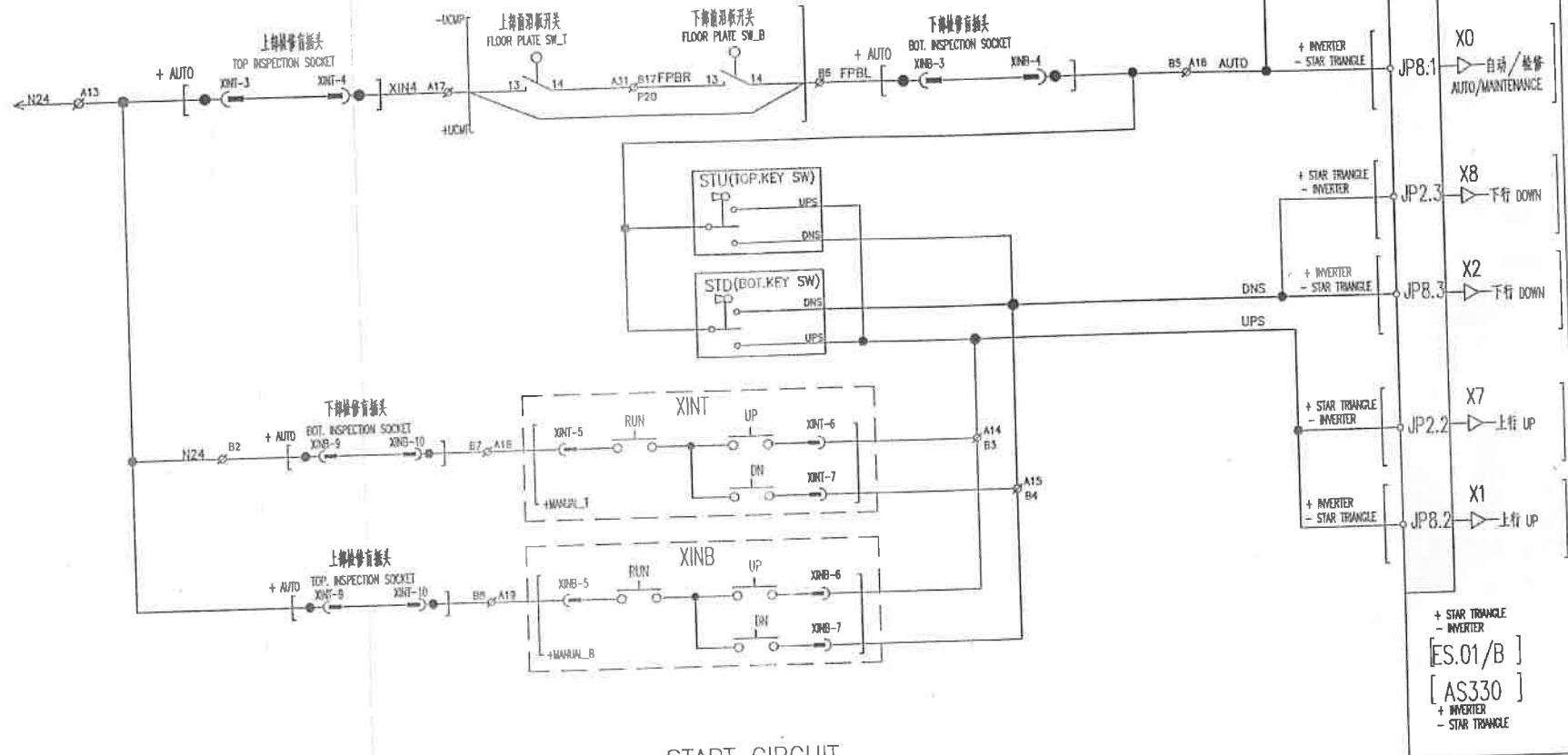
CHANGE  
개 정 안

图纸确认章  
22 MAY 2017

COL 3



FIRE SHUTTER



| NO. | REV. | DATE       | BY   | CHKD | APPD |
|-----|------|------------|------|------|------|
| 1   | 1    | 2017.05.10 | H.FF |      |      |

HYUNDAI  
ELEVATOR CO., LTD.

표준  
제도  
설계  
검토  
승인

임시  
H.FF

보류  
2017.05.10

UNIT  
단위  
MM.  
SCALE  
척도  
1/1  
3RD ANGLE  
3 각 법  
REF. NO  
참고도면

APPAR.  
MCU

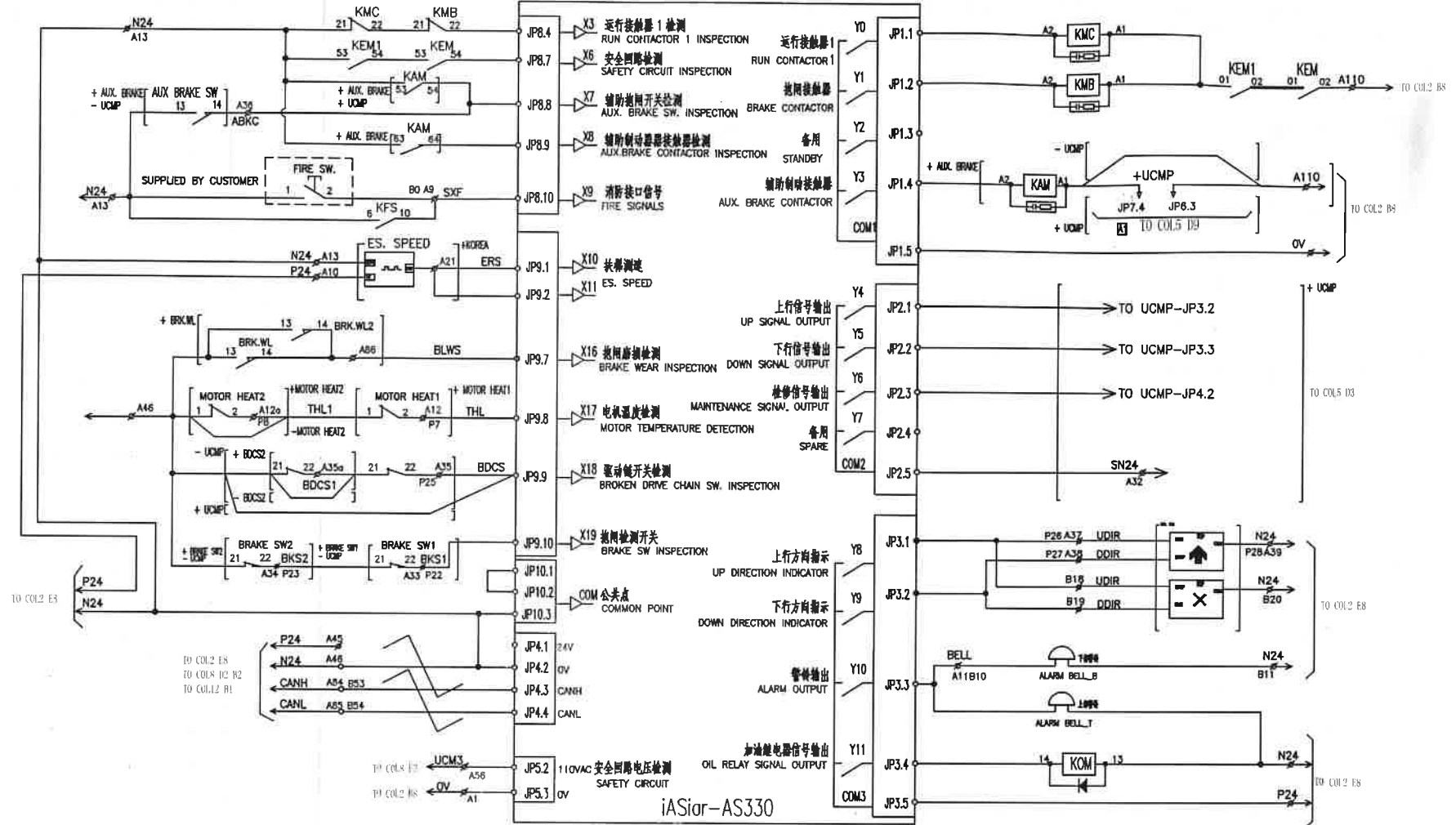
TITLE  
제 목  
CONTROL CIRCUIT SCHEMATIC

DWG. NO  
도면번호  
CHE000057

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이그림이 복사할 수 없음

COL 4

NOTE: THIS DRAWING APPLIED TO INVERTER.



iASior-AS330

CHANGE  
개정판

H.F.F 2017.10.20

| ITEM | NO. | DATE       | DESCRIPTION    |
|------|-----|------------|----------------|
| 1    | 1   | 2017.05.10 | INITIAL DESIGN |
| 2    | 2   | 2017.05.10 | REVISION       |
| 3    | 3   | 2017.05.10 | REVISION       |
| 4    | 4   | 2017.05.10 | REVISION       |
| 5    | 5   | 2017.05.10 | REVISION       |
| 6    | 6   | 2017.05.10 | REVISION       |
| 7    | 7   | 2017.05.10 | REVISION       |
| 8    | 8   | 2017.05.10 | REVISION       |
| 9    | 9   | 2017.05.10 | REVISION       |
| 10   | 10  | 2017.05.10 | REVISION       |

HYUNDAI  
ELEVATOR CO. LTD.

표준  
DRWN BY  
H.F.F  
2017.05.10

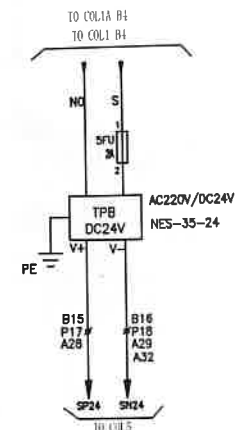
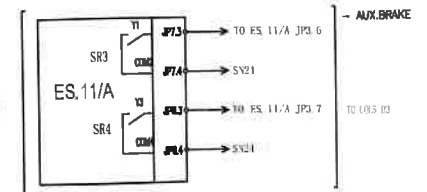
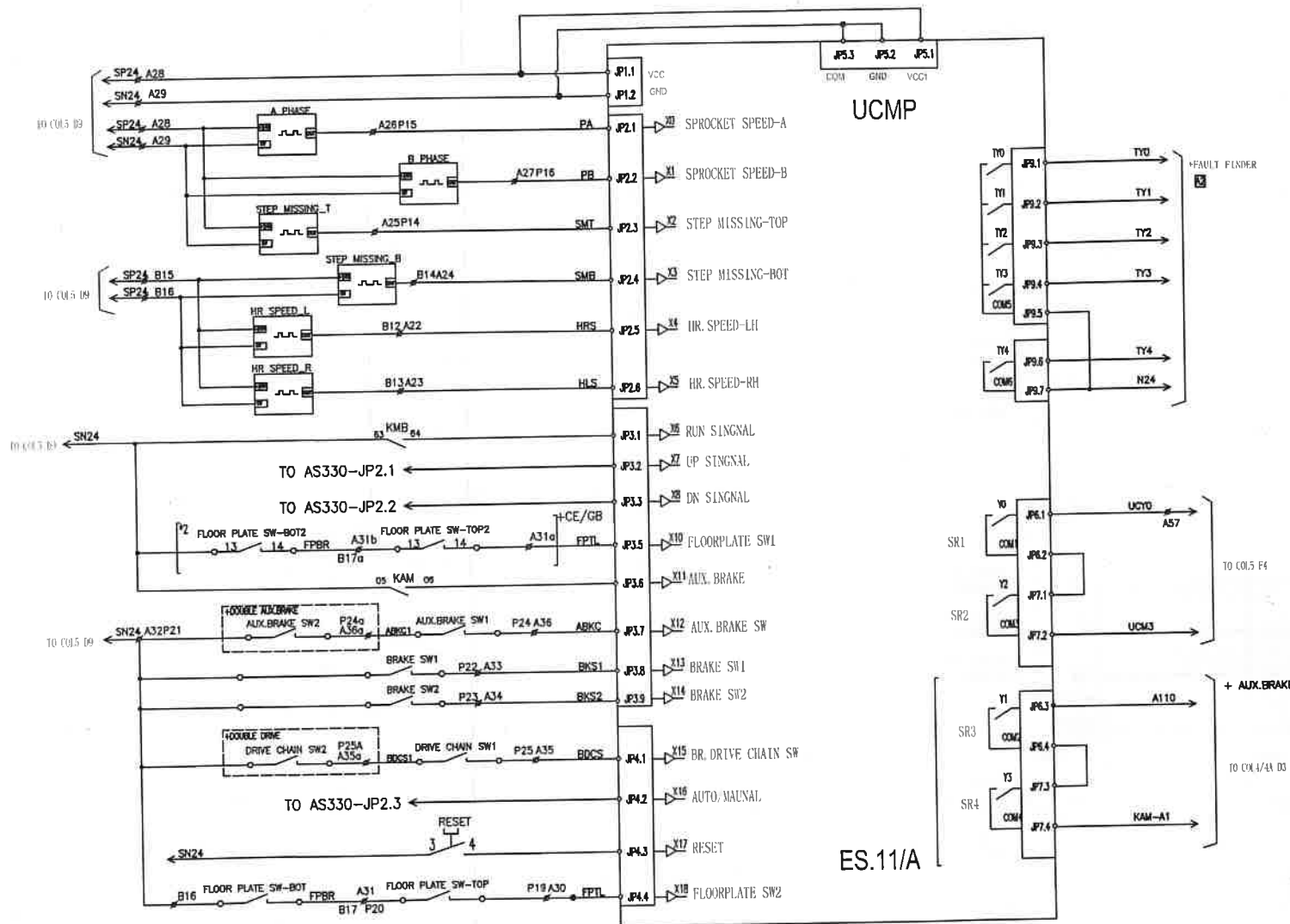
UNIT  
단위  
SCALE  
척도  
3RD ANGLE  
3 각 법  
REF. NO  
참고도면

APPAR. MCU  
TITLE  
제 목  
CONTROL CIRCUIT SCHEMATIC  
DWG. NO  
도면번호  
CHE000057  
4/19

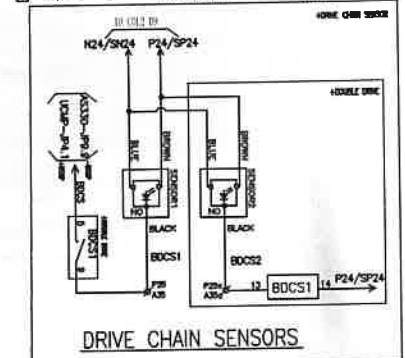
THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이그림이 복사될 수 없음

COL 5

NOTE: THIS DRAWING APPLIED TO INVERTER.



WITH DRIVE CHAIN DETECTION DEVICE IS THE SENSORS.



| REV. | DATE       | DESCRIPTION |
|------|------------|-------------|
| 1    | 2017.05.10 | INITIAL     |
| 2    | 2017.05.10 | REVISION    |
| 3    | 2017.05.10 | REVISION    |
| 4    | 2017.05.10 | REVISION    |
| 5    | 2017.05.10 | REVISION    |
| 6    | 2017.05.10 | REVISION    |
| 7    | 2017.05.10 | REVISION    |
| 8    | 2017.05.10 | REVISION    |
| 9    | 2017.05.10 | REVISION    |
| 10   | 2017.05.10 | REVISION    |

CHANGE  
개정  
DRIVE CHAIN SENSORS(E.B)  
TYO—TY4 (B.7)  
H.FF 2017.10.20

| REV. | DATE       | DESCRIPTION |
|------|------------|-------------|
| 1    | 2017.05.10 | INITIAL     |
| 2    | 2017.05.10 | REVISION    |
| 3    | 2017.05.10 | REVISION    |
| 4    | 2017.05.10 | REVISION    |
| 5    | 2017.05.10 | REVISION    |
| 6    | 2017.05.10 | REVISION    |
| 7    | 2017.05.10 | REVISION    |
| 8    | 2017.05.10 | REVISION    |
| 9    | 2017.05.10 | REVISION    |
| 10   | 2017.05.10 | REVISION    |

**HYUNDAI**  
ELEVATOR CO., LTD.

표준  
DRWN BY  
H.FF  
2017.05.10

DSND BY  
설계  
검토  
승인

CHKD BY  
검토  
승인

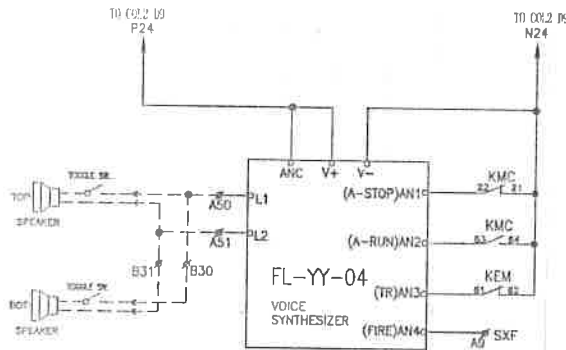
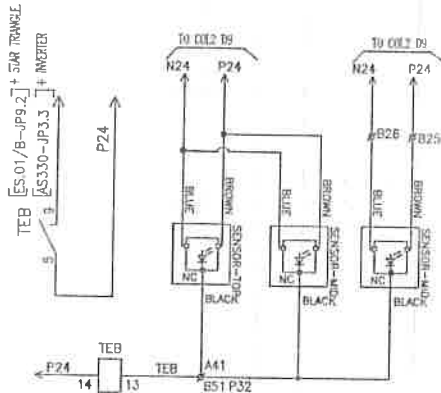
APPD BY  
승인

| UNIT      | MM.   | APPAR.  | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   |       |         | 5/10                      |



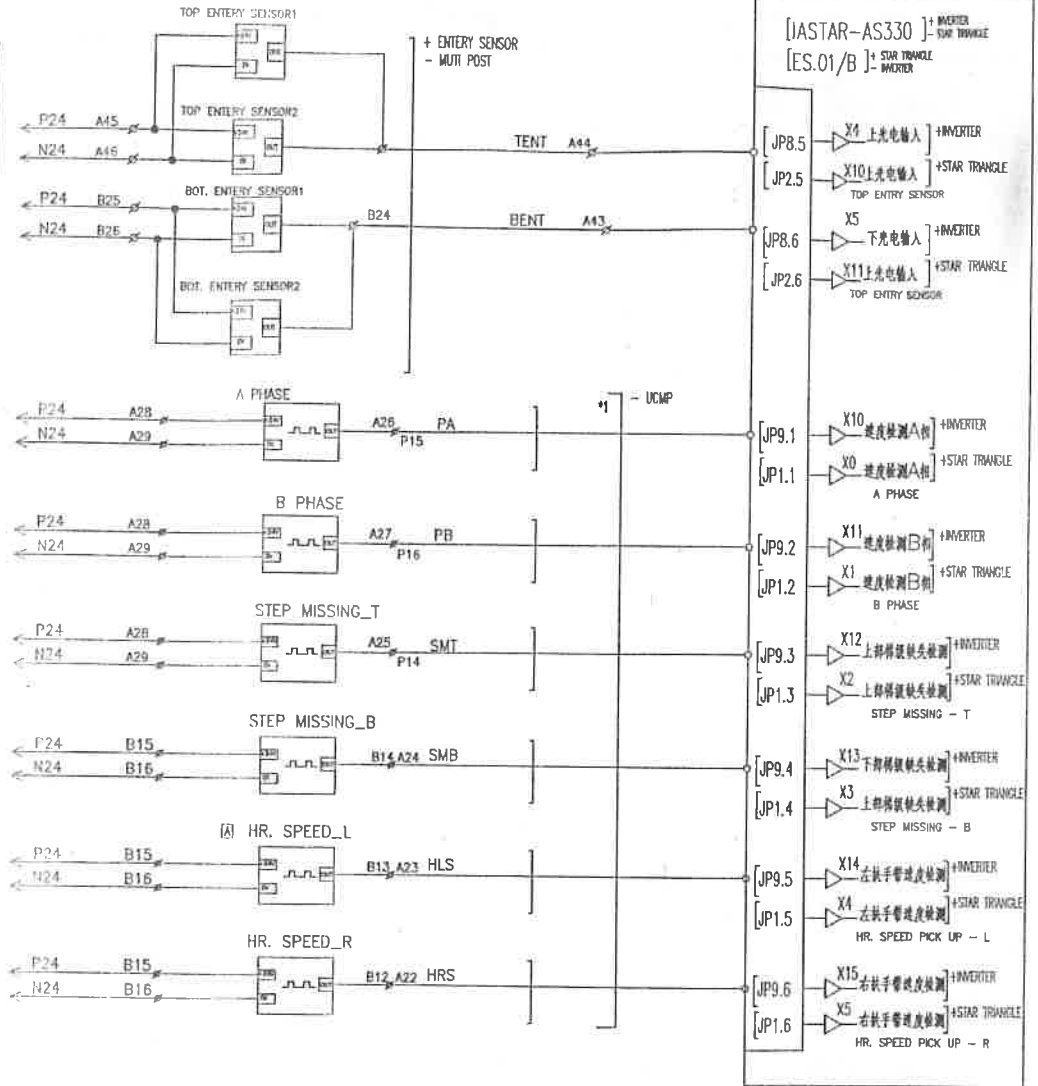
THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO.  
이그리미 복사할 수 없음

COL 6



VOICE SYNTHESIZER

NOTE:  
\*1: WITHOUT UCOMP, P24 IS CONNECTED TO A28 AND B15, N24 IS CONNECTED TO A29 AND B16.



CHANGE  
FL 2017.05.28

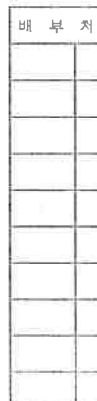
| NO. | REV. | DATE       | BY    | CHKD. | APPD. |
|-----|------|------------|-------|-------|-------|
| 1   | 1    | 2017.05.28 | H.F.F |       |       |
| 2   | 2    | 2017.05.28 | H.F.F |       |       |
| 3   | 3    | 2017.05.28 | H.F.F |       |       |
| 4   | 4    | 2017.05.28 | H.F.F |       |       |
| 5   | 5    | 2017.05.28 | H.F.F |       |       |
| 6   | 6    | 2017.05.28 | H.F.F |       |       |
| 7   | 7    | 2017.05.28 | H.F.F |       |       |
| 8   | 8    | 2017.05.28 | H.F.F |       |       |
| 9   | 9    | 2017.05.28 | H.F.F |       |       |
| 10  | 10   | 2017.05.28 | H.F.F |       |       |

| NO. | REV. | DATE       | BY    | CHKD. | APPD. |
|-----|------|------------|-------|-------|-------|
| 1   | 1    | 2017.05.28 | H.F.F |       |       |
| 2   | 2    | 2017.05.28 | H.F.F |       |       |
| 3   | 3    | 2017.05.28 | H.F.F |       |       |
| 4   | 4    | 2017.05.28 | H.F.F |       |       |
| 5   | 5    | 2017.05.28 | H.F.F |       |       |
| 6   | 6    | 2017.05.28 | H.F.F |       |       |
| 7   | 7    | 2017.05.28 | H.F.F |       |       |
| 8   | 8    | 2017.05.28 | H.F.F |       |       |
| 9   | 9    | 2017.05.28 | H.F.F |       |       |
| 10  | 10   | 2017.05.28 | H.F.F |       |       |

| UNIT      | MM.   | APPAR.  | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1     | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면  | 도면번호    |                           |







|                                                                                                                            |  |  |  |  |                 |         |                 |                           |
|----------------------------------------------------------------------------------------------------------------------------|--|--|--|--|-----------------|---------|-----------------|---------------------------|
|  <b>HYUNDAI</b><br>ELEVATOR CO., LTD. |  |  |  |  | UNIT<br>단위      | MM.<br> | APPAR.<br>      | MCU<br>                   |
| SCALE<br>척도                                                                                                                |  |  |  |  | 1 / 1           |         | TITLE<br>제목     | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE<br>3 각 법                                                                                                         |  |  |  |  | REF. NO<br>참고도면 |         | DWG. NO<br>도면번호 | CHE000057                 |
| 8 / 19                                                                                                                     |  |  |  |  |                 |         |                 |                           |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이/그림이 복사될 수 없음

배 부 적

CHANGE  
개 정 관

图纸确认章  
22 MAY 2017

| REV. | DATE       | BY   | CHKD BY | APPD BY | REMARKS |
|------|------------|------|---------|---------|---------|
| 1    | 2017.05.10 | H.FF |         |         |         |
| 2    | 2017.05.10 | H.FF |         |         |         |
| 3    | 2017.05.10 | H.FF |         |         |         |
| 4    | 2017.05.10 | H.FF |         |         |         |
| 5    | 2017.05.10 | H.FF |         |         |         |
| 6    | 2017.05.10 | H.FF |         |         |         |
| 7    | 2017.05.10 | H.FF |         |         |         |
| 8    | 2017.05.10 | H.FF |         |         |         |
| 9    | 2017.05.10 | H.FF |         |         |         |
| 10   | 2017.05.10 | H.FF |         |         |         |

**HYUNDAI**  
ELEVATOR CO., LTD.

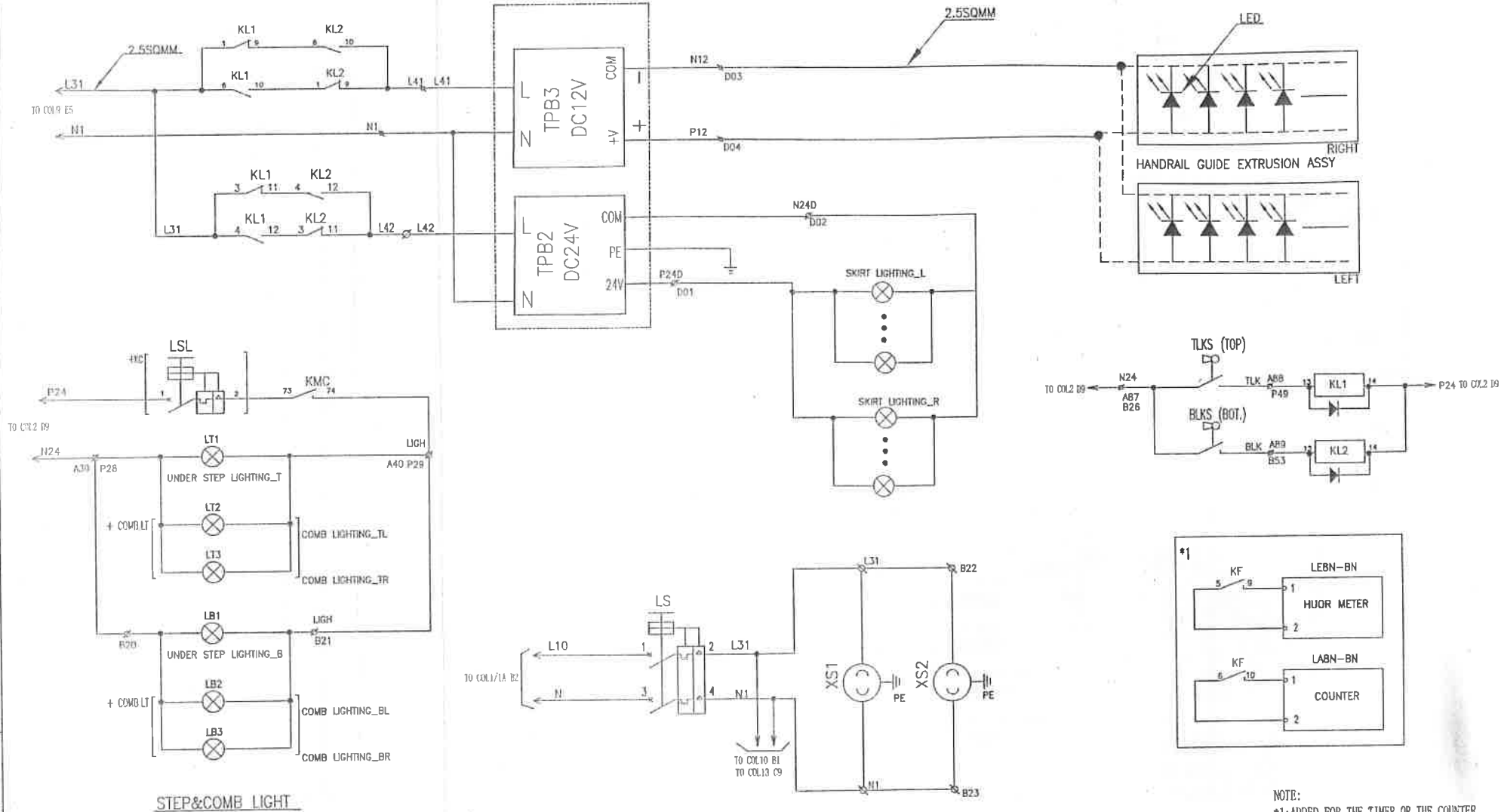
|    |            |         |         |         |
|----|------------|---------|---------|---------|
| 표준 | DRWN BY    | DSND BY | CHKD BY | APPD BY |
| 제도 | 설계         | 검토      | 승인      |         |
| 임시 | H.FF       |         |         |         |
| 보류 | 2017.05.10 |         |         |         |

|           |       |
|-----------|-------|
| UNIT      | MM.   |
| 단위        | mm.   |
| SCALE     | 1/1   |
| 3RD ANGLE | 3 각 법 |
| REF. NO   |       |

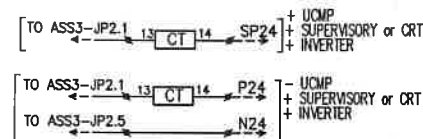
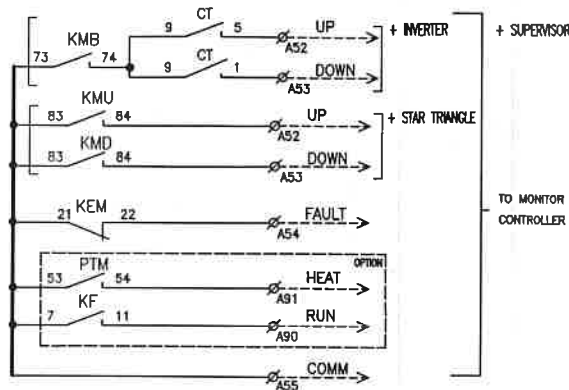
|         |                           |
|---------|---------------------------|
| APPAR.  | MCU                       |
| TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 제 목     |                           |
| DWG. NO | CHE0000057                |

NOTE:  
\*1: ADDED FOR THE TIMER OR THE COUNTER

COL 9

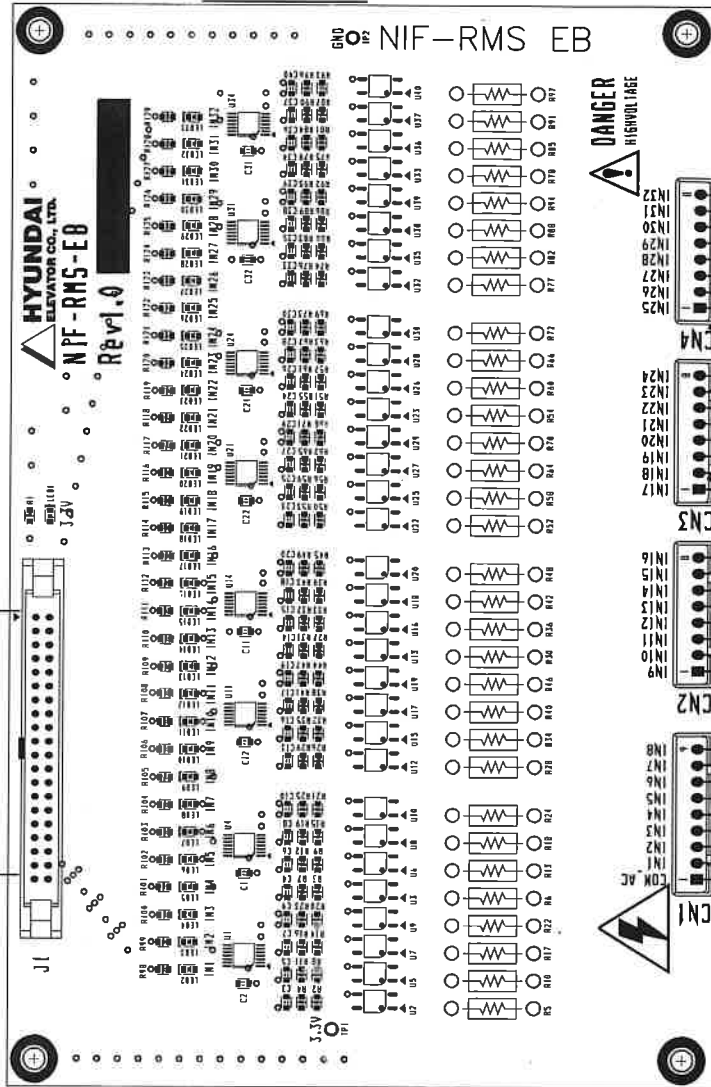


THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이 그림은 헬코의 재산입니다

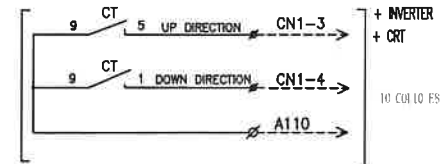
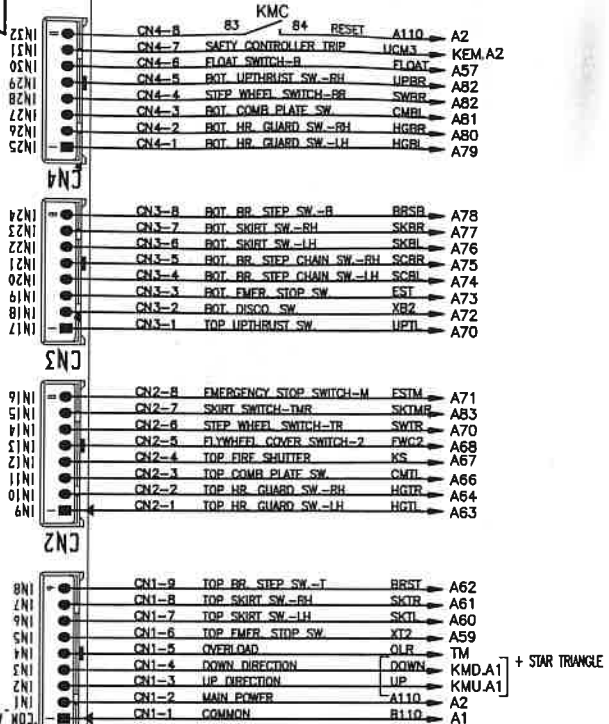


TO  
NIF-RMS

CRT SUPERVISORY BOARD



**DANGER**  
HIGH VOLTAGE



CHANGE  
개정

| REV. | DATE       | BY    | CHKD. | APPD. | REMARKS |
|------|------------|-------|-------|-------|---------|
| 1    | 2017.05.10 | H.F.F |       |       |         |

**HYUNDAI**  
ELEVATOR CO. LTD.

표현  
제도  
설계  
검토  
승인  
H.F.F  
2017.05.10

| UNIT  | MM. | APPAR. | TITLE                     | DWG. NO   | 도면번호  |
|-------|-----|--------|---------------------------|-----------|-------|
| SCALE | 1/1 | MCU    | CONTROL CIRCUIT SCHEMATIC | CHE000057 | 10/19 |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
여기요이 복사할 수 없음

도 수 2

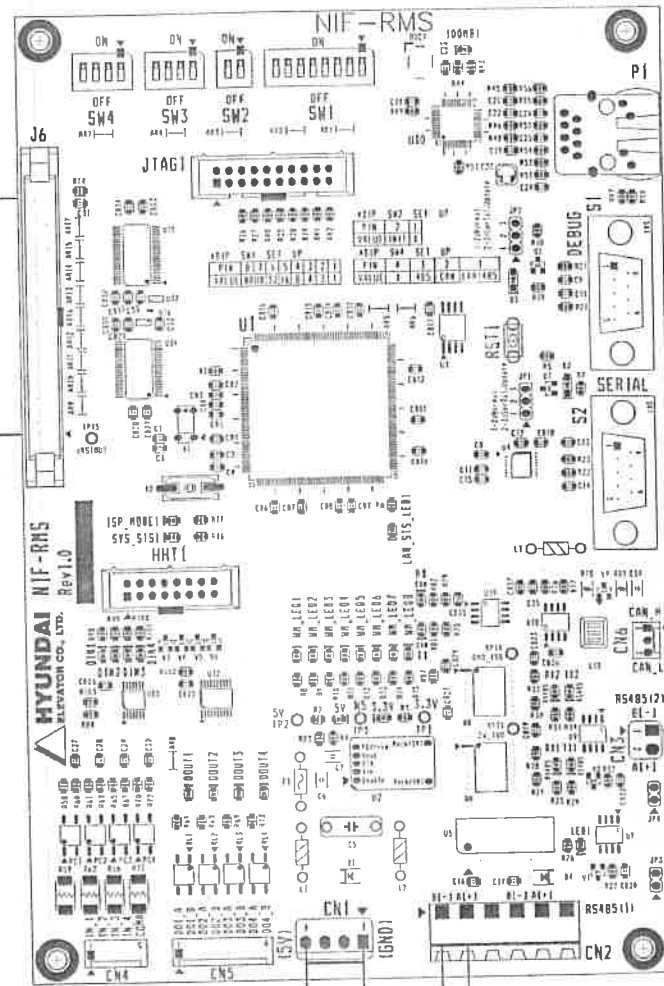
CHANGE  
관심자

图纸确认章  
22 MAY 2017

COL 11

TO  
NIF-RMS EB

CRT COMMUNICATION BOARD



DC5V RS-485

DC5V

N1  
L31

TO CN9 E5

POWER  
SUPPLY

POWER SUPPLY(DC5V)

| TELEPHONE-MASS OPERATOR SPECIFIED |      |      |      |      | UNIT      | MM. | APPAR.   | MCU                       |
|-----------------------------------|------|------|------|------|-----------|-----|----------|---------------------------|
| NO.                               | TURN | BLDG | BLDG | BLDG | SCALE     | 1   | TITLE    | CONTROL CIRCUIT SCHEMATIC |
| NO.                               | NO.  | NO.  | NO.  | NO.  | 1         | 1   | 제목       |                           |
| 1                                 | 1    | 1    | 1    | 1    | 3RD ANGLE | 3   |          |                           |
| 2                                 | 2    | 2    | 2    | 2    | 각도        | 각   |          |                           |
| 3                                 | 3    | 3    | 3    | 3    | REF. NO.  |     | DWG. NO. | CHE000057                 |
| 4                                 | 4    | 4    | 4    | 4    |           |     |          |                           |
| 5                                 | 5    | 5    | 5    | 5    |           |     |          |                           |
| 6                                 | 6    | 6    | 6    | 6    |           |     |          |                           |
| 7                                 | 7    | 7    | 7    | 7    |           |     |          |                           |
| 8                                 | 8    | 8    | 8    | 8    |           |     |          |                           |
| 9                                 | 9    | 9    | 9    | 9    |           |     |          |                           |
| 10                                | 10   | 10   | 10   | 10   |           |     |          |                           |

HYUNDAI  
ELEVATOR CO., LTD.

표준  
DRWN BY  
H.F.F.  
2017.05.10

DSND BY  
설계

CHKD BY  
검토

APPD BY  
승인

11/

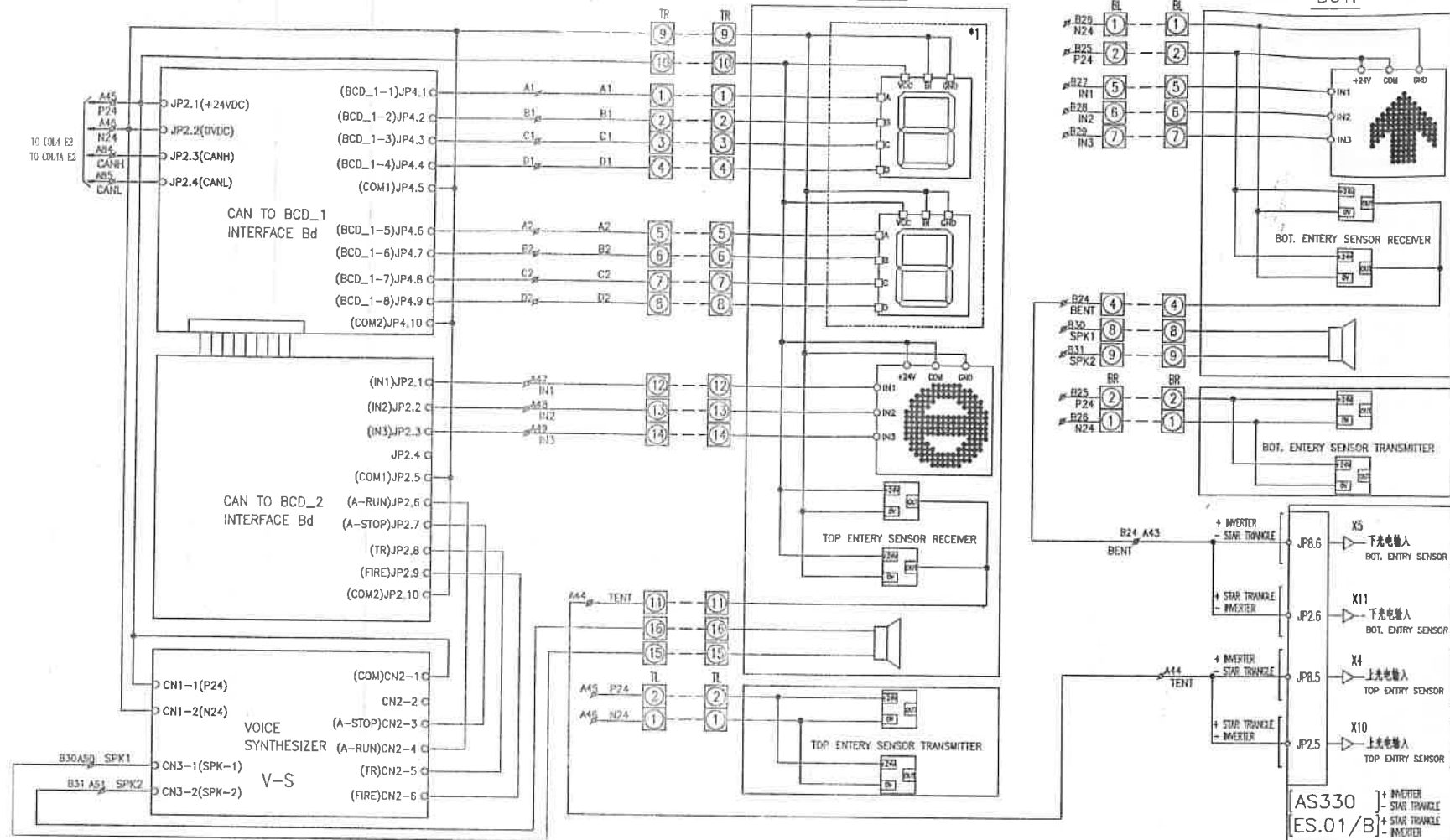
THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
해기없이 복사할 수 없음

| REV | DATE | DESCRIPTION |
|-----|------|-------------|
| 1   |      |             |
| 2   |      |             |
| 3   |      |             |
| 4   |      |             |
| 5   |      |             |
| 6   |      |             |
| 7   |      |             |
| 8   |      |             |
| 9   |      |             |
| 10  |      |             |

COL 12

TOP

BOT.



NOTE:  
\*1: ADDED FOR FAULT DISPLAY WITH MULTI-POST ASP-A/S.

ASP-A/S

图纸确认章  
22 MAY 2017

| REV | DATE | DESCRIPTION |
|-----|------|-------------|
| 1   |      |             |
| 2   |      |             |
| 3   |      |             |
| 4   |      |             |
| 5   |      |             |
| 6   |      |             |
| 7   |      |             |
| 8   |      |             |
| 9   |      |             |
| 10  |      |             |

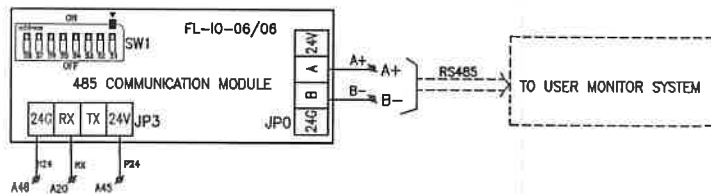
**HYUNDAI**  
ELEVATOR CO., LTD.

|            |         |         |         |   |
|------------|---------|---------|---------|---|
| 표          | 제       | 설       | 검       | 승 |
| DRWN BY    | DSND BY | CHKD BY | APPD BY |   |
| H.F.F      |         |         |         |   |
| 2017.05.10 |         |         |         |   |

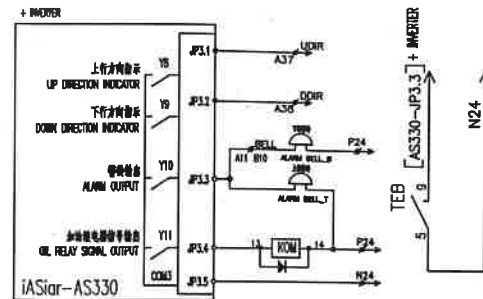
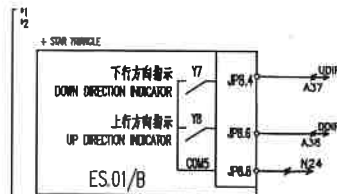
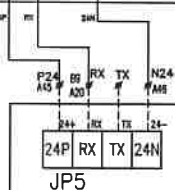
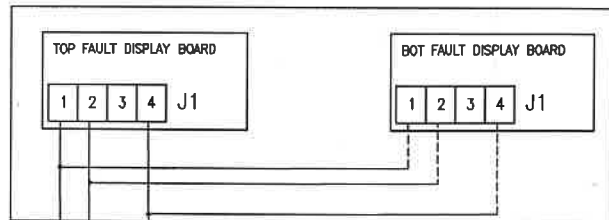
| UNIT      | MM.   | APPAR.  | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 라 법 | DWC. NO | CHE000057                 |
| REF. NO   | 참고도면  | 도면번호    |                           |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
해결이 수 있습니다

COL 13

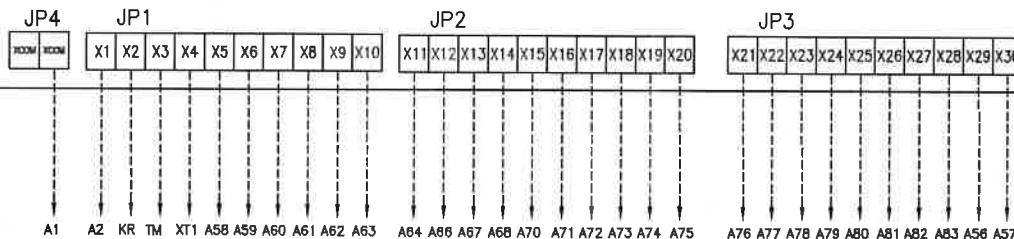


FAULT DISPLAY & DIRECTION INDICATOR IN INNER DECK



FAULT FINDER BOARD

PH-GZB-HD



1 FAULT DISPLAY CODE TABLE

| NO. | SAFETY SWITCH NAME                  | CODE    | REMARK               |
|-----|-------------------------------------|---------|----------------------|
| 1   | FE PROTECTION                       | [01.1]  |                      |
| 2   | MOTOR OVERLOAD ERROR                | [01.2]  |                      |
| 3   | MOTOR THERMOSTAT ERROR              | [01.3]  |                      |
| 4   | PHASE RELAY                         | [01.4]  |                      |
| 5   | CONTROLLER STOP BUTTON              | [01.5]  |                      |
| 6   | TOP STOP BUTTON                     | [01.6]  |                      |
| 7   | TOP LEFT SHORT SW                   | [01.7]  |                      |
| 8   | TOP RIGHT SHORT SW                  | [01.8]  |                      |
| 9   | TOP BROKEN STEP SW                  | [01.9]  |                      |
| 10  | TOP LEFT HILGUARD SW                | [01.10] |                      |
| 11  | TOP RIGHT HILGUARD SW               | [01.11] |                      |
| 12  | TOP COMB PLATE SW                   | [01.12] |                      |
| 13  | FIRE SW ERROR                       | [01.13] |                      |
| 14  | FURNACE COVER SW                    | [11.4]  | OPTION               |
| 15  | TOP UP/HOIST SW                     | [11.5]  | OPTION               |
| 16  | MIDDLE STOP BUTTON                  | [11.6]  | OPTION               |
| 17  | BOT. DISCONNECT STOP BUTTON         | [11.7]  |                      |
| 18  | BOT. STOP BUTTON                    | [11.8]  |                      |
| 19  | BOT. LEFT HIL. STEP CHAIN SW        | [11.9]  |                      |
| 20  | BOT. RIGHT HIL. STEP CHAIN SW       | [11.10] |                      |
| 21  | BOT. LEFT SHORT SW                  | [11.11] |                      |
| 22  | BOT. RIGHT SHORT SW                 | [11.12] |                      |
| 23  | BOT. BROKEN STEP SW                 | [11.13] |                      |
| 24  | BOT. LEFT HIL. GUARD SW             | [11.14] |                      |
| 25  | BOT. RIGHT HIL. GUARD SW            | [11.15] |                      |
| 26  | BOT COMB SW                         | [11.16] |                      |
| 27  | BOT. UP/HOIST SW                    | [11.17] | OPTION               |
| 28  | MIDDLE SHORT SW                     | [11.18] | OPTION               |
| 29  | EARTHQUAKE SENSOR                   | [11.19] | OPTION               |
| 30  | FLOAT SW                            | [11.20] | OPTION               |
| NO. | UCMP FAULT NAME                     | CODE    | REMARK               |
| 31  | REAL-TIME CLOCK BREAKDOWN           | [11.1]  | UCMP INNER FAULT     |
| 32  | REDUNDANCY INSPECTION ERROR         | [11.2]  | UCMP INNER FAULT     |
| 33  | PARAMETER CRC ERROR                 | [11.3]  | UCMP INNER FAULT     |
| 34  | FAULT CRC ERROR                     | [11.4]  | UCMP INNER FAULT     |
| 35  | UNREASONABLE PARAMETER              | [11.5]  | UNREASONABLE SET     |
| 36  | POWER VOLTAGE FAULT                 | [11.6]  | VOLTAGE IS DOWN      |
| 37  | SAFETY RELAY FAULT                  | [11.7]  | UCMP INNER FAULT     |
| 38  | OVERSPEED MORE THAN 120%            | [11.8]  | CHECK THE PARAMETERS |
| 39  | OVERSPEED MORE THAN 140%            | [11.9]  | CHECK THE ENVELOPE   |
| 40  | UNDERSTOCK LOWER THAN 80%           | [11.10] | IS OVERSTOCKED       |
| 41  | AB PHASE REVERSE CONNECTION         | [11.11] | CHECK RELAY BOWLER   |
| 42  | UNUSUAL DROP OF SPEED               | [11.12] | CHECK DRIVE MOTOR    |
| 43  | DRIVE CHAIN DISCONNECTED            | [11.13] | CHECK CHAIN SW IF IT |
| 44  | EXCEEDING THE MAXIMUM STOPPING      | [11.14] | CHECK STOPPING FORCE |
| 45  | FAULT OF AUXILIARY BRAKE            | [11.15] | CHECK RELAY BOWLER   |
| 46  | MISSING OF STEP (UP)                | [11.16] | CHECK WHETHER IT IS  |
| 47  | MISSING OF STEP (DOWN)              | [11.17] | CHECK WHETHER IT IS  |
| 48  | MISSING OF FLOOR PLATE SWITCH       | [11.18] | CHECK WHETHER IT IS  |
| 49  | FAULT OF FLOOR PLATE SWITCH         | [11.19] | CHECK WHETHER THE SW |
| 50  | WORK BRAKE CAN NOT OPENED           | [11.20] | CHECK RELAY BOWLER   |
| 51  | WORK BRAKE NOT CLOSED               | [11.21] | CHECK WHETHER RELAY  |
| 52  | UNDERSTOCK OF LEFT HANDRAIL         | [11.22] | CHECK WHETHER SW IS  |
| 53  | UNDERSTOCK OF LEFT HANDRAIL         | [11.23] | CHECK WHETHER SW IS  |
| 54  | UNDERSTOCK OF RIGHT HANDRAIL        | [11.24] | CHECK WHETHER SW IS  |
| 55  | OVERSTOCK OF RIGHT HANDRAIL         | [11.25] | CHECK WHETHER SW IS  |
| 56  | OPENING MANUAL RELEASEMENT WAS      | [11.26] | CHECK WHETHER IT IS  |
| 57  | SLIP DURING LEFT-STOP               | [11.27] | CHECK WHETHER IT IS  |
| 58  | THE 3-TO CHANNELS BE NOT CORRESPOND | [11.28] | CHECK THE CODE       |
| 59  | LOSS OF AB PHASE                    | [11.29] | CHECK THE CODE       |

\*1: 485-100 FOR FAULT DISPLAY WITH DIRECTION INDICATOR.  
\*2: 485-100 FOR MULTI-POST ASP-A/S.  
\*3: 485-100 FOR FAULT DISPLAY WITH MULTI-POST ASP-A/S.

HYUNDAI  
ELEVATOR CO., LTD

표준  
원시  
보류  
DRWN BY  
제 도  
설 계  
DSND BY  
검 토  
CHKD BY  
승 인  
H.F.F  
2017.05.10

UNIT  
단위  
SCALE  
척도  
JRD ANGLE  
3 각 법  
REF. NO  
DWG. NO

APPAR.  
MCU  
TITLE  
제목  
CONTROL CIRCUIT SCHEMATIC  
DWG. NO

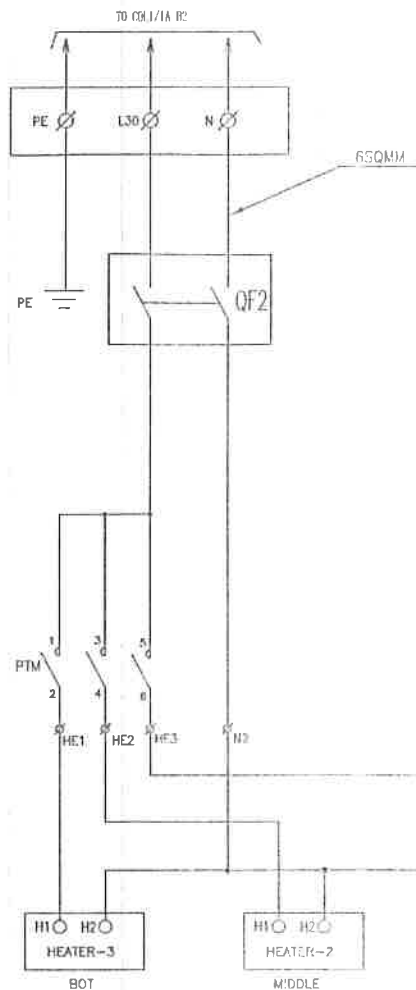
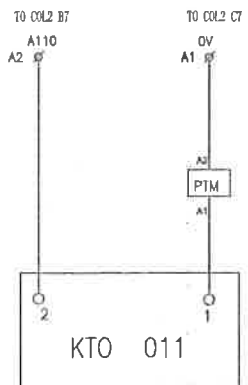


THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이 도면은 헬코의 소유이며  
복사할 수 없음

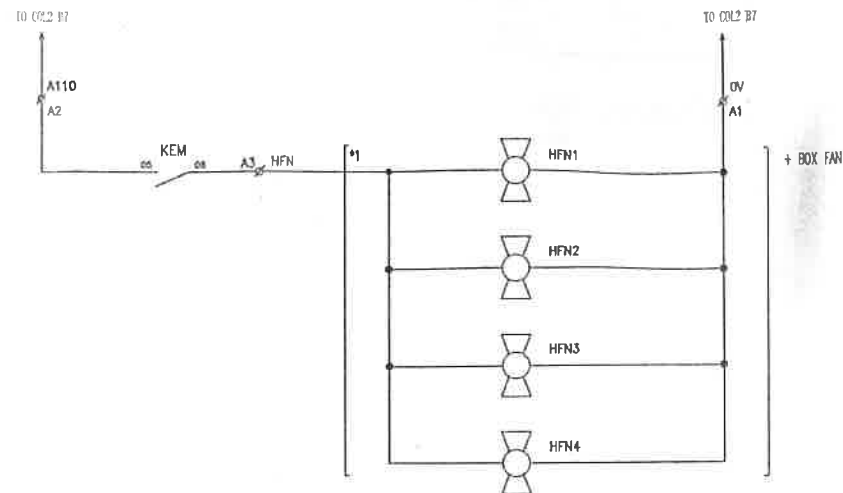
변경 이력

CHANGE  
개정 이력

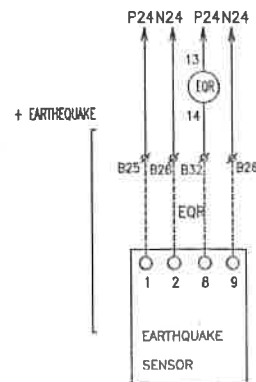
+ HEATER



COL 14



NOTE:  
\*1: IF THERE IS A RESISTANCE BOX, USE THE FAN.



| NO. | DATE       | BY   | CHKD BY | APPD BY |
|-----|------------|------|---------|---------|
| 1   | 2017.05.10 | H.FF | 5/17    | 5/19    |

| DATE       | BY   | CHKD BY | APPD BY |
|------------|------|---------|---------|
| 2017.05.10 | H.FF | 5/17    | 5/19    |

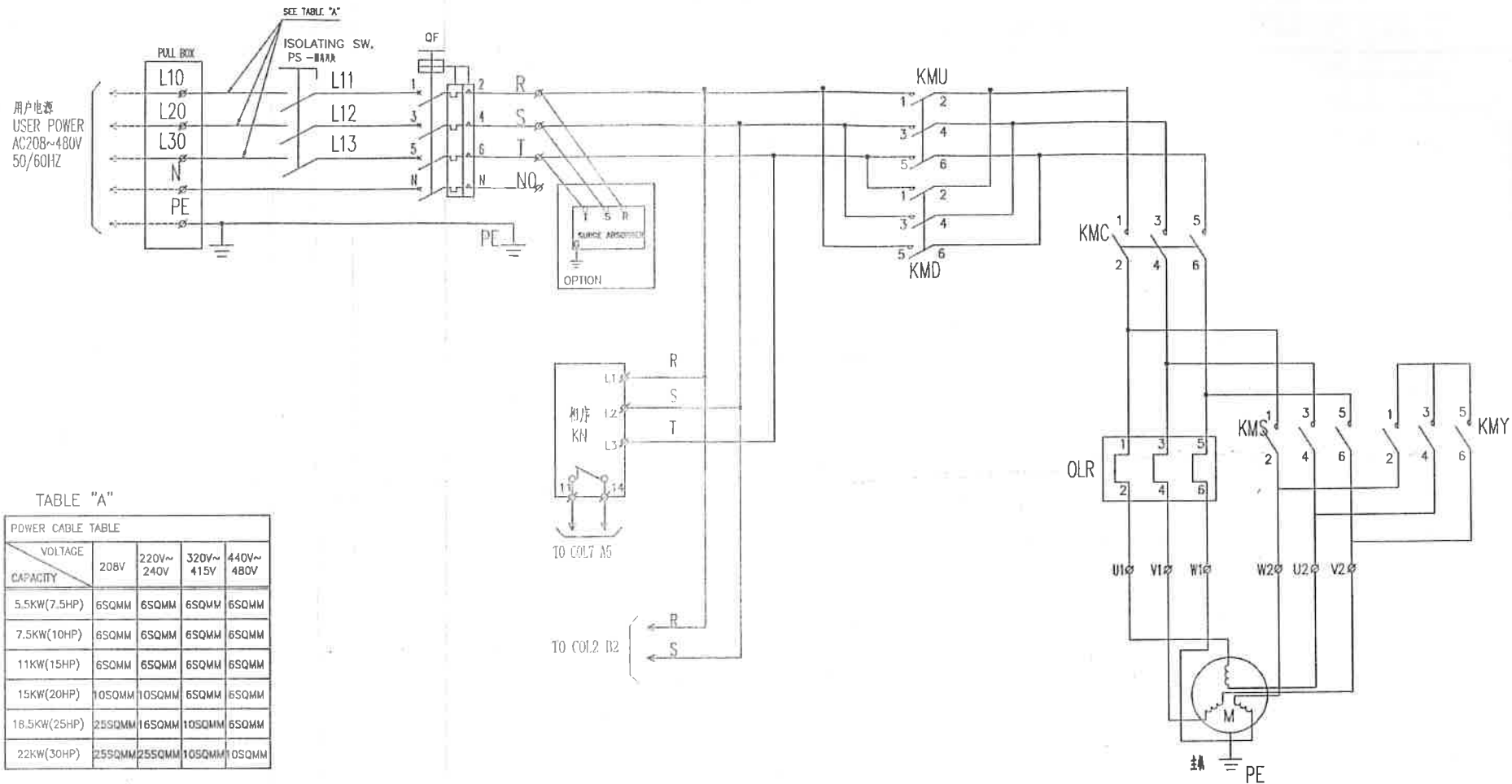
| UNIT      | MM.   | APPR.   | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면  | 도면번호    | 14/19                     |





COL 1A

NOTE: THIS DRAWING APPLIED TO STAR TRIANGLE.



NOTE:  
\*1: ADDED FOR DOUBLE DRIVE

图纸确认章  
22 MAY 2017

| SIZE | DATE | REV. | BY | CHKD BY | APPD BY |
|------|------|------|----|---------|---------|
| 1    | 18   | 01   | 01 | 01      | 01      |
| 1    | 18   | 02   | 02 | 02      | 02      |
| 1    | 18   | 03   | 03 | 03      | 03      |
| 1    | 18   | 04   | 04 | 04      | 04      |
| 1    | 18   | 05   | 05 | 05      | 05      |
| 1    | 18   | 06   | 06 | 06      | 06      |
| 1    | 18   | 07   | 07 | 07      | 07      |
| 1    | 18   | 08   | 08 | 08      | 08      |
| 1    | 18   | 09   | 09 | 09      | 09      |
| 1    | 18   | 10   | 10 | 10      | 10      |
| 1    | 18   | 11   | 11 | 11      | 11      |
| 1    | 18   | 12   | 12 | 12      | 12      |
| 1    | 18   | 13   | 13 | 13      | 13      |
| 1    | 18   | 14   | 14 | 14      | 14      |
| 1    | 18   | 15   | 15 | 15      | 15      |
| 1    | 18   | 16   | 16 | 16      | 16      |
| 1    | 18   | 17   | 17 | 17      | 17      |
| 1    | 18   | 18   | 18 | 18      | 18      |
| 1    | 18   | 19   | 19 | 19      | 19      |
| 1    | 18   | 20   | 20 | 20      | 20      |

HYUNDAI  
ELEVATOR CO., LTD.

| 표준 | DRWN BY    | DSND BY | CHKD BY | APPD BY |
|----|------------|---------|---------|---------|
| 제도 | 제도         | 제도      | 제도      | 제도      |
| 임시 | H.F.F      |         |         |         |
| 보류 | 2017.05.10 |         |         |         |

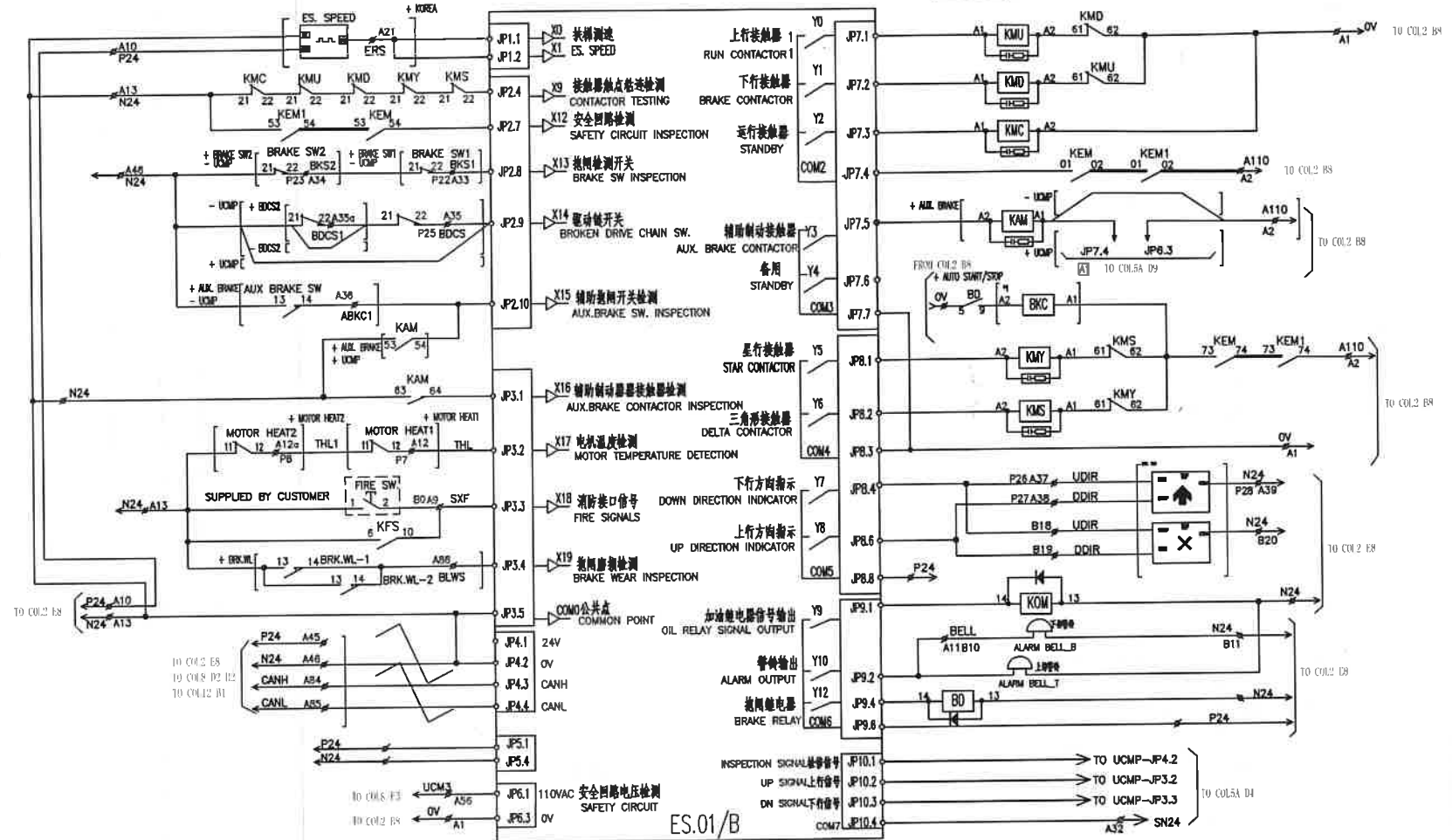
| UNIT      | MM. |
|-----------|-----|
| 단위        | 1   |
| SCALE     | 1/1 |
| 척도        | 1/1 |
| 3RD ANGLE |     |
| 3 각 법     |     |
| REF. NO   |     |
| 참고도면      |     |

| APPAR.  | MCU                       |
|---------|---------------------------|
| TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 제목      |                           |
| DWG. NO | CHE000057                 |
| 도면번호    |                           |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
이것이 헬코의 재산일 수 있음

COL 4A

NOTE: THIS DRAWING APPLIED TO STAR TRIANGLE.



NOTE:  
\*1:USED IF PROVIDED WITH MULTI-POST OR AUTO START/STOP(EXCEPT CHINA).

제 목

CHANGE  
개 정 연

DATE: 2017.10.20  
H.F.F

| NO. | DATE       | REVISION | DESCRIPTION    |
|-----|------------|----------|----------------|
| 1   | 2017.05.10 | 1        | INITIAL DESIGN |
| 2   | 2017.05.10 | 2        | REVISION       |
| 3   | 2017.05.10 | 3        | REVISION       |
| 4   | 2017.05.10 | 4        | REVISION       |
| 5   | 2017.05.10 | 5        | REVISION       |
| 6   | 2017.05.10 | 6        | REVISION       |
| 7   | 2017.05.10 | 7        | REVISION       |
| 8   | 2017.05.10 | 8        | REVISION       |
| 9   | 2017.05.10 | 9        | REVISION       |
| 10  | 2017.05.10 | 10       | REVISION       |

HYUNDAI  
ELEVATOR CO., LTD.

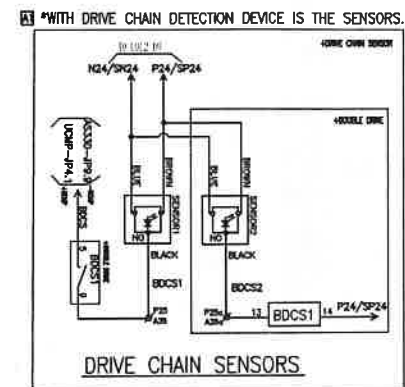
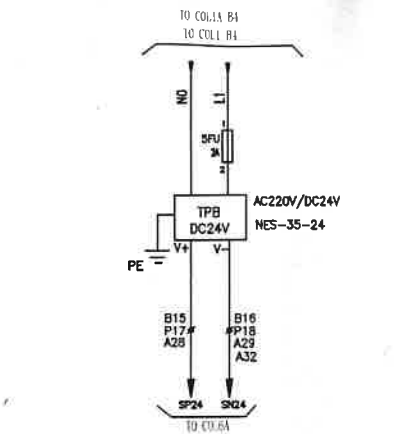
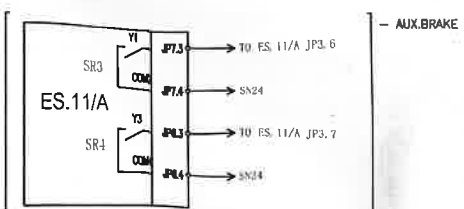
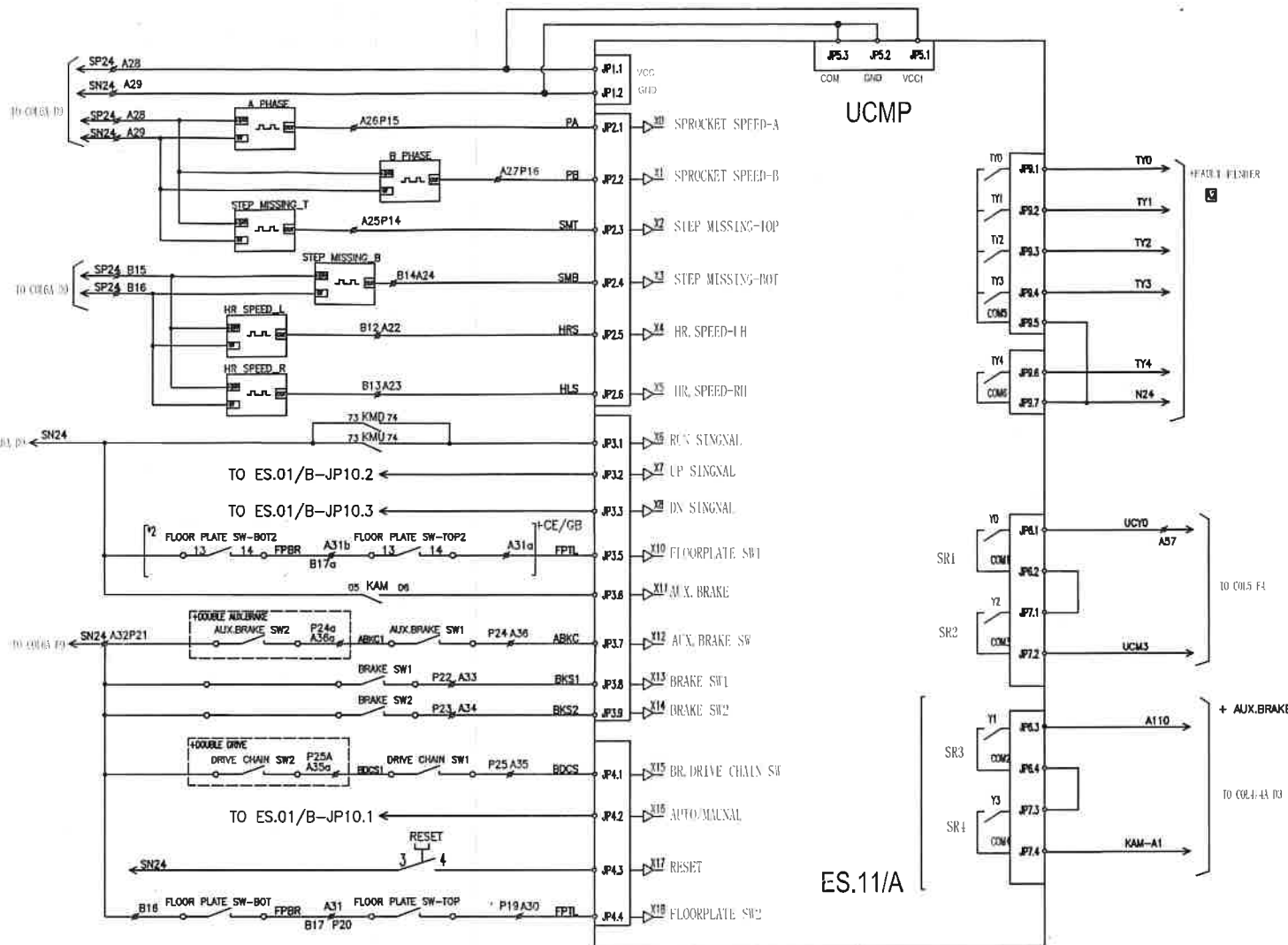
표 준  
제 도  
설 계  
검 토  
승 인  
H.F.F  
2017.05.10

| UNIT      | MM.   | APPAR.  | MCU                       |
|-----------|-------|---------|---------------------------|
| SCALE     | 1/1   | TITLE   | CONTROL CIRCUIT SCHEMATIC |
| 3RD ANGLE | 3 각 법 | DWG. NO | CHE000057                 |
| REF. NO   | 참고도면  | 도면번호    |                           |

THIS DRAWING IS PROPRIETARY TO HELLCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELLCO  
허가없이 복사할 수 없음

COL 5A

NOTE: THIS DRAWING APPLIED TO STAR TRIANGLE.



| REV | DATE       | DESCRIPTION |
|-----|------------|-------------|
| 1   | 2017.05.10 | INITIAL     |
| 2   | 2017.05.10 | REVISION    |
| 3   | 2017.05.10 | REVISION    |
| 4   | 2017.05.10 | REVISION    |
| 5   | 2017.05.10 | REVISION    |
| 6   | 2017.05.10 | REVISION    |
| 7   | 2017.05.10 | REVISION    |
| 8   | 2017.05.10 | REVISION    |
| 9   | 2017.05.10 | REVISION    |
| 10  | 2017.05.10 | REVISION    |

CHANGE  
계정  
ADDED DRIVE CHAIN SENSORS(E.S.)  
USED TYO--TY4 (B.7)  
H.F.F 2017.10.20

| SIZE | UNIT | QTY | REMARKS |
|------|------|-----|---------|
| 1    | mm   | 1   | 1       |
| 2    | mm   | 1   | 1       |
| 3    | mm   | 1   | 1       |
| 4    | mm   | 1   | 1       |
| 5    | mm   | 1   | 1       |
| 6    | mm   | 1   | 1       |
| 7    | mm   | 1   | 1       |
| 8    | mm   | 1   | 1       |
| 9    | mm   | 1   | 1       |
| 10   | mm   | 1   | 1       |

| DRWN BY    | DSND BY | CHKD BY | APPD BY |
|------------|---------|---------|---------|
| H.F.F      |         |         |         |
| 2017.05.10 |         |         |         |

| UNIT                      | SCALE | 3RD ANGLE | REF. NO | 참고도면 |
|---------------------------|-------|-----------|---------|------|
| mm                        | 1/1   | 각각        |         |      |
| APPAR.                    | MCU   |           |         |      |
| TITLE                     | 제 목   |           |         |      |
| CONTROL CIRCUIT SCHEMATIC |       |           |         |      |
| DWG. NO                   | 도면번호  |           |         |      |
| CHE000057                 |       |           |         |      |

THIS DRAWING IS PROPRIETARY TO HELCO  
NO PART OF THIS DRAWING MAY BE REPRODUCED  
WITHOUT THE PERMISSION OF HELCO  
여기서부터 복사할 수 없음

# COL 16

## INDEX & LABEL LIST

| ITEM                       | SYMBOL | REMARK       | ITEM          | SYMBOL | REMARK       |
|----------------------------|--------|--------------|---------------|--------|--------------|
| ISOLATING SWITCH           |        |              | COL           |        |              |
| BREAKER                    |        |              | CONNECTOR     |        |              |
| CONNECTOR                  |        |              | KEY SW.       |        |              |
| A CONTACT                  |        | MAIN CONTACT | BUTTON        |        |              |
| FUSE                       |        |              | OUTPUT SIGNAL |        |              |
| TERMINAL BLOCK             |        |              | SENSOR        |        |              |
| MOTOR ROOM HEAD PUSHBUTTON |        |              | B CONTACT     |        | AUX. CONTACT |
| STOP BUTTON                |        |              | A CONTACT     |        | AUX. CONTACT |
| RELAY & CONTACTOR COIL     |        |              | THREE LINE    |        |              |
|                            |        |              | FAN           |        |              |
|                            |        |              | RECEPTACLE    |        |              |
|                            |        |              | EARTH         |        |              |
|                            |        |              | BELL          |        |              |
|                            |        |              | MAGNET RING   |        |              |

| LABEL      | COL NO.    | LABEL DESCRIPTION        |
|------------|------------|--------------------------|
| OF         | COL1/COL1A | AIR BREAK SWITCH         |
| KN         | COL1/COL1A | PHASE DETECTOR           |
| KR         | COL1A      | OVERLOAD PROTECTOR       |
| RB         | COL1       | INVERTER BRAKE RESISTOR  |
| TCO        | COL2       | TRANSFORMER              |
| PS         | COL2       | ISLATING SWITCH          |
| CPS/TPB    | COL2/COL5  | POWER SWITCH             |
| FU/FA      | COL2       | FUSE                     |
| BK         | COL2       | BRAKE                    |
| ABK        | COL2       | AUX.BRAKE                |
| OIL        | COL2       | OIL                      |
| XINT       | COL3       | TOP INSPECTION SOCKET    |
| XINB       | COL3       | BOT.INSPECTION SOCKET    |
| KF/KFS/SHR | COL3/COL2  | RELAY                    |
| STU/STD    | COL3       | TOP/BOT.KEY SWITCH       |
| SXF        | COL4/COL4A | FIRE                     |
| THL        | COL4/COL4A | MOTOR THERMISTOR         |
| ERS        | COL4/COL4A | SPEED SENSOR             |
| KMU/KMD    | COL4A      | UP/DOWN CONTACTOR        |
| KEM        | COL7       | SAFETY CIRCUIT CONTACTOR |
| BELL       | COL4/COL4A | BELL                     |
| KAM        | COL4/COL4A | AUX.BRAKE CONTACTOR      |
| KMB        | COL4       | BRAKE CONTACTOR          |
| BKC        | COL4A      | BRAKE CONTACTOR          |
| BD         | COL4A      | BRAKE RELAY              |
| KOM        | COL4/COL4A | OIL RELAY                |
| UCMP       | COL5/COL5A | SAFETY MONITORING BOARD  |
| PA/PB      | COL5/COL5A | A/B PHASE                |
| SMT/SMB    | COL5/COL5A | TOP/BOT.STEP MISSING     |
| HRS/HLS    | COL5/COL5A | HANDRAIL SPEED SENSOR    |
| RESET      | COL5/COL5A | RESET BUTTON             |

| LABEL          | COL NO.    | LABEL DESCRIPTION       |
|----------------|------------|-------------------------|
| ABKC           | COL5/COL5A | AUX.BRAKE SWITCH        |
| BKS            | COL5/COL5A | BRAKE SWITCH            |
| FPBR/FPTL      | COL5/COL5A | FLOOR PLATE SWITCH      |
| FL-YY-04       | COL6       | VOICE SYNTHESIZER       |
| UDIR/DDIR      | COL4/COL4A | DIRECTION INDICATOR     |
| TEB            | COL6       | BUZZER SIGNAL SENSOR    |
| TENT/BENT      | COL6       | TOP/BOT.ENTRY SENSOR    |
| ASP            | COL12      | MULTI-POST              |
| TLKS/BLKS      | COL9       | LIGHTING KEY SWITCH     |
| KL1/KL2        | COL9       | LIGHTING RELAY          |
| LS             | COL9       | AIR SWITCH              |
| XS1/XS2        | COL9       | TOP/BOT. SOCKET         |
| HUOR METER     | COL9       | HUOR METER              |
| COUNTER        | COL9       | COUNTER                 |
| PH-GZB-HD      | COL13      | FAULT FINDER BOARD      |
| FDB            | COL13      | FAULT DISPLAY BOARD     |
| NIF-RMS EB     | COL10      | CRT SUPERVISORY BOARD   |
| NIF-RMS        | COL11      | CRT COMMUNICATION BOARD |
| KTO            | COL14      | TEMPERATURE CONTROLLER  |
| HEATER         | COL14      | HEATER                  |
| HR.HEATER      | COL14      | HANDRAIL HEATER         |
| HFN            | COL14      | BOX FAN                 |
| iASior-AS330   | COL1/COL4  | INVERTER                |
| ES.01/B        | COL4A      | MAIN BAORD              |
| SURGE ABSORBER | COL1/COL1A | SURGE ABSORBER          |
| UCMP           | COL5/COL5A | SAFETY MONITOR BAORD    |
| ES.02/F        | COL8       | SAFETY MONITOR BAORD    |

图纸确认章  
22 MAY 2017

HYUNDAI  
ELEVATOR CO., LTD.

표준  
H.F.F  
2017.05.10

UNIT  
SCALE  
3RD ANGLE  
REF. NO

APPAR. MCU  
TITLE CONTROL CIRCUIT SCHEMATIC  
DWG. NO CHE000057

H. FF 2018-10-11

~~20/22~~



S-BT ESCALATOR SCHEMATIC/ WIRING DIAGRAM  
MARCH 2019  
Revision 11

## **Hyundai Elevator Co.,Ltd.**

### **Head office & Factory**

San136-1, Ami-ri,Bubal-eud,Ichon-shi,Kyonggi-do,Korea

P.C: 467734

Tel: 82-31-644-5114

Fax: 82-2-745-4227

### **Seoul Office**

1-83 Dongsoong-dong,Jongro-ku,Seoul,Korea

P.C: 11050

Tel: 82-2-3670-0660

Fax: 82-2-3672-8763/4

## **Shanghai Hyundai Elevator Mfg.Co.,Ltd.**

### **Head office & Factory**

No.182 Zhenxi (South)Liantang Town,Qinpu District,  
Shanghai,China

P.C: 201716

Tel: (021) 5981-3900

Fax: (021) 5981-3903

### **Shanghai Office**

Room 505-507 5 fl.,Yuanzhong Cientific Research Bldg.,  
No.1905 Hongmei Road, Shanghai,China

P.C: 200233

Tel: (021) 6485-8600/7030

Fax: (021) 6485-4953

### **Beijing Office**

Room#518,Gaolan Bldg.,No.32 Liangmaqiao Road  
Chaoyang District,Beijing,China

P.C: 100004

Tel: (010) 6464-3509

Fax: (010) 6464-3394